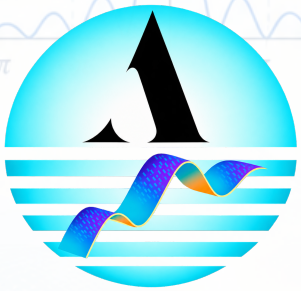
"The purpose of learning mathematics is not merely to find answers, but to understand the ideas behind them."



Fourier Series and Integral Transforms

Fundamentals and Computational Techniques

Lecture Notes in Undergraduate Mathematics

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Study Materials for UG Mathematics, PG Mathematics and Entrance Examinations –

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CHAPTER
1

Fourier Series

1.1 Introduction to Fourier Series

Fourier analysis provides a fundamental framework for representing functions in terms of trigonometric components and studying their frequency characteristics. Fourier series express periodic functions as infinite sums of sines and cosines, revealing the relationship between periodicity and harmonic structure. These representations play an important role in mathematical analysis, physics, engineering, and signal processing. This chapter introduces periodic functions, Fourier coefficients, and the fundamental principles of Fourier series, together with their properties and applications.

Learning objectives. After completing this chapter, the reader should be able to identify periodic functions and their symmetries; derive and calculate Fourier coefficients for periods 2π and $2L$; determine sums of Fourier series at continuous and discontinuous points; construct half-range cosine and sine expansions; and use Bessel's inequality, the Riemann–Lebesgue lemma, and Parseval's identity.

How to study this chapter. In each worked example, begin by identifying the period and interval, test the symmetry of the function, calculate the required coefficients, and only then discuss convergence. A formula for the coefficients answers how to construct the series; the convergence theorem answers what the series represents at a particular point.

1.1.1 Periodic Functions

Periodic functions arise naturally in the mathematical description of oscillations, waves, and other phenomena that repeat at regular intervals. Their recurring structure is fundamental to Fourier analysis, which represents suitable periodic functions in terms of trigonometric components.

Definition 1.1.1 (Periodic Function)**.** Let $D \subseteq \mathbb{R}$, and let $f : D \to \mathbb{R}$ be a function. A number $T > 0$ is called a *period* of f if $x + T \in D$ whenever $x \in D$ and $f(x + T) = f(x)$ for all $x \in D$. The function is called *periodic* if it has at least one positive period. If the set of positive periods has a least element, that element is called the *fundamental period* of f.

Example 1.1.2.

1. The sine function has fundamental period 2π, since $\sin(x + 2\pi) = \sin x$ for every $x \in \mathbb{R}$. Its positive integer multiples $4\pi, 6\pi, \ldots$ are also periods.
2. The cosine function likewise has fundamental period 2π.
3. The tangent function has fundamental period π, because $\tan(x + \pi) = \tan x$ wherever it is defined.
4. Every $T > 0$ is a period of a constant function $f(x) = c$ on $\mathbb{R}$. Consequently, a constant function has no fundamental period.

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Properties of Periodic Functions

Let f and g be periodic functions defined on $\mathbb{R}$.

1. **Linear combinations.** If T is a common period of f and g, then $c_1 f + c_2 g$ also has period T, for any constants $c_1, c_2 \in \mathbb{R}$.
2. **Products and quotients.** If T is a common period, then fg has period T. The quotient f/g is periodic with period T on its natural domain, provided $g(x) \neq 0$ at the points where the quotient is evaluated.
3. **Horizontal scaling.** If T is a period of f and $k \neq 0$, then $T/|k|$ is a period of $f(kx)$. For example, $\sin(3x)$ has fundamental period $2\pi/3$.
4. **Commensurable periods.** If T_1 and T_2 are periods of f and g, respectively, and T_1/T_2 is rational, they possess a common positive integer multiple T. Consequently, $f + g$ is periodic with period T. The least such common multiple is a period, though it need not be the fundamental period of the sum.

1.1.2 Even and Odd Functions

The symmetry of a function often simplifies its Fourier coefficients by causing entire families of terms to vanish. We recall the following definitions.

Definition 1.1.3 (Even Function)**.** A function $f : D \to \mathbb{R}$, defined on a domain symmetric about the origin, is called *even* if $f(-x) = f(x)$ for every $x \in D$. Its graph is symmetric about the y-axis.

Examples include x^2, x^4, $\cos x$, and $|x|$.

Definition 1.1.4 (Odd Function)**.** A function $f : D \to \mathbb{R}$, defined on a domain symmetric about the origin, is called *odd* if $f(-x) = -f(x)$ for every $x \in D$. Its graph is symmetric about the origin.

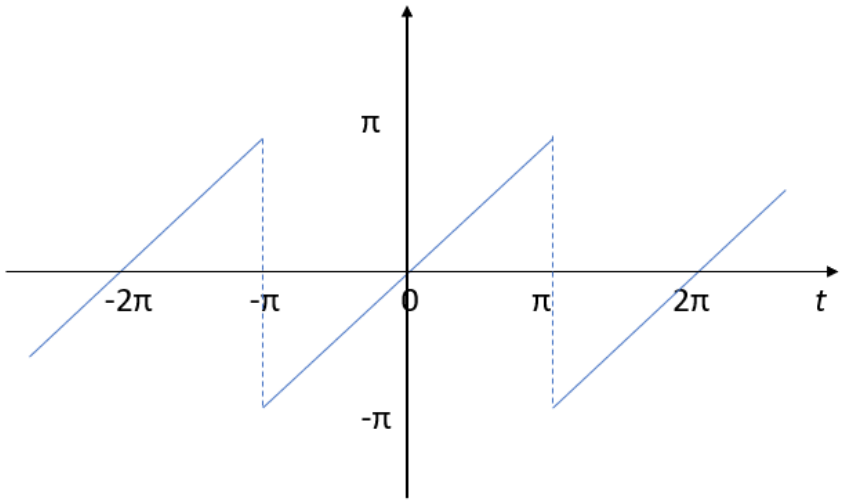
Examples include x, x^3, $\sin x$, and $\tan x$ on their respective domains.

Properties of Even and Odd Functions

The following elementary properties are particularly useful in Fourier analysis.

1. **Algebraic combinations.**
   - The product of two even functions or two odd functions is even.
   - The product of an even function and an odd function is odd.
   - The sum or difference of two even functions is even; the sum or difference of two odd functions is odd.
2. **Even–odd decomposition.** Every function f on a domain symmetric about the origin can be decomposed uniquely as
$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even part}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd part}}.$$
3. **Integration over symmetric intervals.** If f is integrable on $[-L, L]$, then $\int_{-L}^{L} f(x)\,dx = 0$ when f is odd, whereas $\int_{-L}^{L} f(x)\,dx = 2\int_{0}^{L} f(x)\,dx$ when f is even.

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1.1.3 Special Periodic Waveforms

Square, sawtooth, and triangular waves provide standard examples of nonsinusoidal periodic signals. Each can be represented by a Fourier series, illustrating how complex waveforms arise from superpositions of elementary trigonometric functions. We introduce them here with period 2π.

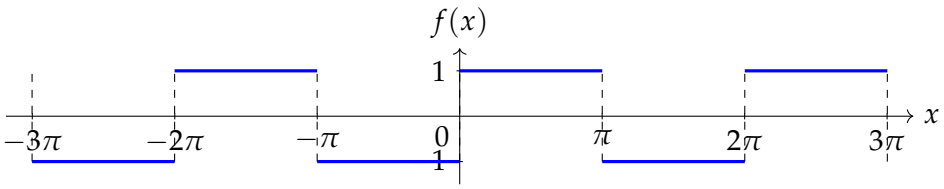
Square Wave

A square wave alternates between two constant levels and has jump discontinuities at regular intervals.

Definition 1.1.5. The standard square wave of period 2π is defined on $(-\pi, \pi)$ by

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$

and extended periodically by $f(x + 2\pi) = f(x)$. Values at the points of discontinuity may be left unspecified. Wherever both sides are defined, this square wave is odd: $f(-x) = -f(x)$.



Sawtooth Wave

A sawtooth wave increases linearly over each period and then returns abruptly to its initial level.

Definition 1.1.6. Define $f(x) = x$ for $-\pi < x < \pi$ and extend the function periodically with period 2π. The resulting sawtooth wave is odd wherever it is defined and exhibits jump discontinuities at odd multiples of π.

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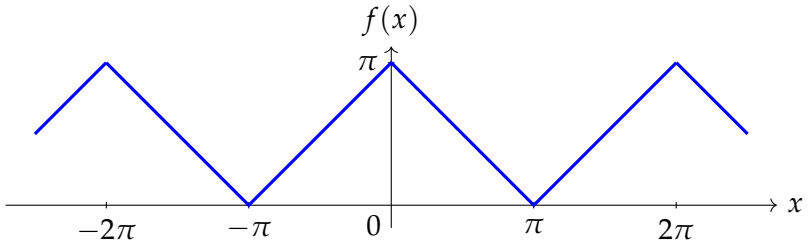
Triangular Wave

A triangular wave consists of alternating linear segments and is continuous, although its slope changes abruptly at the vertices.

Definition 1.1.7. The triangular wave of period 2π is given by $f(x) = \pi - |x|$ on $[-\pi, \pi]$ and extended periodically. Equivalently,

$$f(x) = \begin{cases} \pi + x, & -\pi \le x < 0, \\ \pi - x, & 0 \le x \le \pi. \end{cases}$$

Since $f(-x) = f(x)$, the triangular wave is even.



1.2 The Fourier Series Representation

The basic idea of Fourier series is to describe a periodic function in terms of sines and cosines of increasing frequencies. Under suitable hypotheses, these trigonometric terms reproduce the function at its points of continuity and yield the average of the one-sided limits at a jump discontinuity.

1.2.1 Fourier Series for Functions of Period $2L$

Let $L > 0$ and suppose f is a $2L$-periodic, integrable function. Its Fourier series is constructed from the trigonometric system $1, \cos(n\pi x/L)$, and $\sin(n\pi x/L)$, for $n \ge 1$.

Definition 1.2.1 (Fourier Series)**.** The *Fourier series* associated with f is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right),$$

where a_0, a_n, and b_n are the Fourier coefficients. The symbol $\sim$ denotes the formal Fourier expansion and does not, by itself, assert pointwise convergence.

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1.2.2 Euler's Formulae for the Fourier Coefficients

The Fourier coefficients are determined by the orthogonality of sine and cosine functions over $[-L, L]$. For nonnegative integers m, n, the relevant identities are

$$\int_{-L}^{L} \cos\frac{m\pi x}{L}\cos\frac{n\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ L, & m = n \geq 1, \\ 2L, & m = n = 0, \end{cases}$$

$$\int_{-L}^{L} \sin\frac{m\pi x}{L}\sin\frac{n\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ L, & m = n \geq 1, \\ 0, & m = n = 0, \end{cases}$$

$$\int_{-L}^{L} \cos\frac{m\pi x}{L}\sin\frac{n\pi x}{L}\,dx = 0.$$

The coefficient formulas can be obtained by multiplying a suitable Fourier expansion by a trigonometric basis function and integrating over one period. This calculation is valid, for example, when the series converges uniformly; more generally, the coefficients are defined directly by the integrals below.

Calculating the Coefficients

- **Constant coefficient.** Integrating over $[-L, L]$ eliminates the nonconstant sine and cosine terms, giving $\int_{-L}^{L} f(x)\,dx = La_0$.
- **Cosine coefficients.** For $n \geq 1$, multiply by $\cos(n\pi x/L)$ and integrate over one period. Orthogonality leaves only the term containing a_n:

$$\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx = La_n.$$

- **Sine coefficients.** Multiplication by $\sin(n\pi x/L)$ gives, for $n \geq 1$,

$$\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx = Lb_n.$$

Theorem 1.2.2 (Euler's Formulae). *Let f be an integrable function of period $2L$, where $L > 0$. Its Fourier coefficients are*

$$\begin{aligned} a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)\,dx, \\ a_n &= \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx, \qquad n \geq 1, \\ b_n &= \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx, \qquad n \geq 1. \end{aligned}$$

Proof. The Fourier series associated with f is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{L} + b_n\sin\frac{n\pi x}{L}\right).$$

We use the orthogonality relations

$$\int_{-L}^{L} \cos\frac{m\pi x}{L}\cos\frac{n\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ L, & m = n \geq 1, \end{cases}$$

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$$\int_{-L}^{L} \sin\frac{m\pi x}{L}\sin\frac{n\pi x}{L}\,dx = \begin{cases} 0, & m \neq n, \\ L, & m = n \geq 1, \end{cases}$$

and

$$\int_{-L}^{L} \sin\frac{m\pi x}{L}\cos\frac{n\pi x}{L}\,dx = 0$$

for all positive integers m, n.

Suppose first that the trigonometric expansion converges to f in the mean-square sense (or uniformly), so that taking its inner products with fixed trigonometric functions is justified. Integrating the Fourier expansion over $[-L, L]$, all sine and cosine terms vanish. Hence

$$\int_{-L}^{L} f(x)\,dx = \frac{a_0}{2}(2L) = a_0 L,$$

which gives $a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$.

Next, multiplying the expansion by $\cos(m\pi x/L)$ and integrating over $[-L, L]$, orthogonality gives

$$\int_{-L}^{L} f(x)\cos\frac{m\pi x}{L}\,dx = a_m L.$$

Therefore,

$$a_m = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{m\pi x}{L}\,dx.$$

Similarly, multiplying by $\sin(m\pi x/L)$ and integrating, we obtain

$$\int_{-L}^{L} f(x)\sin\frac{m\pi x}{L}\,dx = b_m L,$$

and consequently,

$$b_m = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{m\pi x}{L}\,dx.$$

Since $m \geq 1$ is arbitrary, the required formulas follow. For an arbitrary integrable f, the integral expressions define its Fourier coefficients independently of any prior convergence assumption. ■

1.2.3 Fourier Series for Functions of Period 2π

A very common and important case is when the period of the function is $T = 2\pi$. This corresponds to setting $L = \pi$ in the general formulas. The term $\frac{n\pi x}{L}$ simplifies to $\frac{n\pi x}{\pi} = nx$.

The Fourier series for a function $f(x)$ with period 2π on the interval $[-\pi, \pi]$ is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right)$$

The corresponding Euler's formulae for the coefficients become:

$$\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \quad (n = 1, 2, 3, \dots) \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \quad (n = 1, 2, 3, \dots) \end{aligned}$$

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Remark 1.2.3 (Interval of Integration). The interval of integration must have a length equal to the period $T = 2L$. While $[-L, L]$ is common, any interval such as $[0, 2L]$ or $[c, c + 2L]$ can be used. For a period of 2π, the interval $[0, 2\pi]$ is frequently used instead of $[-\pi, \pi]$. The choice often depends on the definition of the function.

Example 1.2.4 (Fourier Series of a Square Wave)**.** Find the Fourier series for the square wave function $f(x)$ with period $T = 2\pi$ defined on $(-\pi, \pi)$ as:

$$f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$$

Solution. The function is periodic with period 2π, so we set $L = \pi$. By inspection, $f(-x) = -f(x)$, so the function is odd. Consequently, all cosine coefficients are zero: $a_n = 0$ for $n \geq 0$. We need only compute the sine coefficients, b_n. Since $f(x)\sin(nx)$ is an even function (a product of two odd functions), we can write:

$$\begin{aligned} b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx = \frac{2}{\pi}\int_0^{\pi}(1)\sin(nx)\,dx \\ &= \frac{2}{\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi} = \frac{2}{n\pi}(1-\cos(n\pi)) \end{aligned}$$

The term $(1 - \cos(n\pi))$ is 0 if n is even, and 2 if n is odd. Thus, $b_n = 0$ for even n and $b_n = \frac{4}{n\pi}$ for odd n. The resulting Fourier series is composed only of sine terms with odd harmonics:

$$f(x) \sim \sum_{n=1,3,5,\ldots}^{\infty} \frac{4}{n\pi}\sin(nx) = \frac{4}{\pi}\left(\sin(x) + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + \ldots\right)$$

■

Example 1.2.5 (Fourier Series of a Triangular Wave)**.** Find the Fourier series for the triangular wave function $f(x) = \pi - |x|$ with period $T = 2\pi$ on the interval $[-\pi, \pi]$.

Solution. The function has period 2π ($L = \pi$) and is an even function since $f(-x) = \pi - |-x| = \pi - |x| = f(x)$. Therefore, all sine coefficients are zero: $b_n = 0$ for all $n \geq 1$. The DC component, a_0, is calculated as:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - |x|)\,dx = \frac{2}{\pi}\int_0^{\pi}(\pi - x)\,dx = \frac{2}{\pi}\left[\pi x - \frac{x^2}{2}\right]_0^{\pi} = \pi$$

For the cosine coefficients a_n ($n \geq 1$), the integrand $(\pi - |x|)\cos(nx)$ is even:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - |x|)\cos(nx)\,dx = \frac{2}{\pi}\int_0^{\pi}(\pi - x)\cos(nx)\,dx$$

Using integration by parts, we find:

$$\begin{aligned} a_n &= \frac{2}{\pi}\left(\left[(\pi - x)\frac{\sin(nx)}{n}\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi}\sin(nx)\,dx\right) \\ &= \frac{2}{n\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi} = \frac{2}{n^2\pi}(1-\cos(n\pi)) = \frac{2}{n^2\pi}(1-(-1)^n) \end{aligned}$$

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This yields $a_n = 0$ for even n and $a_n = \frac{4}{n^2\pi}$ for odd n. The Fourier series is:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1,3,5,\ldots}^{\infty} a_n \cos(nx) = \frac{\pi}{2} + \frac{4}{\pi} \sum_{n=1,3,5,\ldots}^{\infty} \frac{\cos(nx)}{n^2}$$

$$f(x) \sim \frac{\pi}{2} + \frac{4}{\pi}\left(\cos(x) + \frac{1}{9}\cos(3x) + \frac{1}{25}\cos(5x) + \ldots\right)$$

■

The next example illustrates why the choice of integration interval need not be symmetric about the origin. A function specified over $[0, 2]$ is best integrated over that interval. Provided the interval covers exactly one complete period, the resulting Fourier coefficients are unchanged.

Example 1.2.6 (Fourier Series for Period 2L)**.** Find the Fourier series for the function $f(x) = x$ defined on the interval $[0, 2)$ with period $T = 2$.

Solution. The function has period $T = 2$, so $2L = 2$ which implies $L = 1$. We compute the coefficients by integrating over the interval $[0, 2]$. The DC component a_0 is:

$$a_0 = \frac{1}{1}\int_0^2 x\,dx = \left[\frac{x^2}{2}\right]_0^2 = 2$$

The cosine coefficients a_n for $n \geq 1$ are:

$$a_n = \int_0^2 x\cos(n\pi x)\,dx = \left[\frac{x\sin(n\pi x)}{n\pi}\right]_0^2 - \int_0^2 \frac{\sin(n\pi x)}{n\pi}dx = 0 - \left[\frac{\cos(n\pi x)}{n^2\pi^2}\right]_0^2 = 0$$

The sine coefficients b_n for $n \geq 1$ are:

$$\begin{aligned} b_n = \int_0^2 x\sin(n\pi x)\,dx &= \left[\frac{-x\cos(n\pi x)}{n\pi}\right]_0^2 + \int_0^2 \frac{\cos(n\pi x)}{n\pi}dx \\ &= -\frac{2\cos(2n\pi)}{n\pi} + \left[\frac{\sin(n\pi x)}{n^2\pi^2}\right]_0^2 = -\frac{2}{n\pi} \end{aligned}$$

Assembling the series with $a_0 = 2$, $a_n = 0$, and $b_n = -2/(n\pi)$:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} b_n \sin(\frac{n\pi x}{L}) = \frac{2}{2} + \sum_{n=1}^{\infty}\left(-\frac{2}{n\pi}\right)\sin(n\pi x)$$

$$f(x) \sim 1 - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi x)}{n} = 1 - \frac{2}{\pi}\left(\sin(\pi x) + \frac{1}{2}\sin(2\pi x) + \frac{1}{3}\sin(3\pi x) + \ldots\right)$$

■

Exercise 1.2.7. Find the Fourier series expansion for each of the following functions:

1. Find the Fourier series of the square wave function defined by:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi < x < 0 \\ 0 & \text{for } 0 \leq x < \pi, \end{cases} \qquad f(x + 2\pi) = f(x).$$

with period 2π.

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2. Determine the Fourier series of the sawtooth wave function:

$$f(x) = \begin{cases} 2+x & \text{for } -2 \le x \le 0 \\ 2-x & \text{for } 0 \le x \le 2, \end{cases} \qquad f(x+4) = f(x).$$

with period 4.

3. Compute the Fourier series of the triangular wave function:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \le x \le 0 \\ \pi - x & \text{for } 0 \le x \le \pi \end{cases}$$

with period 2π.

For each function:

- Calculate the Fourier coefficients a_0, a_n, and b_n
- Write the complete Fourier series expansion
- State the value to which the series converges at points of discontinuity (if any)

1.3 Fourier Series of Even and Odd Functions

The symmetry properties introduced at the beginning of the chapter now yield a substantial simplification in the actual computation of Fourier coefficients. On a symmetric interval, an odd integrand has integral zero, while the integral of an even integrand can be evaluated over half the interval and doubled. Thus, before calculating coefficients, one should always test whether the function is even, odd, or neither.

For example, an even function multiplied by a sine is odd, so its sine coefficient vanishes. Similarly, an odd function multiplied by a cosine is odd, so all its cosine coefficients vanish. This observation often reduces three sets of coefficient calculations to only one.

Fourier Series Simplifications

For a function $f(x)$ with period $2L$:

- If $f(x)$ is **even**, then all b_n coefficients vanish:

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx = 0$$

The Fourier series becomes a **Fourier cosine series**:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where

$$a_n = \frac{2}{L}\int_{0}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

- If $f(x)$ is **odd**, then all a_n coefficients vanish:

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx = 0$$

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The Fourier series becomes a **Fourier sine series**:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Example 1.3.1 (Even Function). Find the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = (-x)^2 = x^2 = f(x)$, the function is even. We compute the coefficients:

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2 dx = \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{2\pi^2}{3}$$

For $n \geq 1$:

$$a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos(nx)dx$$

Using integration by parts twice:

$$a_n = \frac{2}{\pi}\left[\frac{x^2\sin(nx)}{n} + \frac{2x\cos(nx)}{n^2} - \frac{2\sin(nx)}{n^3}\right]_0^{\pi} = \frac{2}{\pi}\left(\frac{2\pi\cos(n\pi)}{n^2}\right) = \frac{4(-1)^n}{n^2}$$

Thus, the Fourier series is:

$$x^2 \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\frac{4(-1)^n}{n^2}\cos(nx)$$

Example 1.3.2 (Odd Function). Find the Fourier series of $f(x) = x$ on $[-\pi, \pi]$.

Solution: Since $f(-x) = -x = -f(x)$, the function is odd. We compute the coefficients:

$$b_n = \frac{2}{\pi}\int_0^{\pi} x\sin(nx)dx$$

Using integration by parts:

$$b_n = \frac{2}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^{\pi} = \frac{2}{\pi}\left(-\frac{\pi\cos(n\pi)}{n}\right) = \frac{2(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$x \sim \sum_{n=1}^{\infty}\frac{2(-1)^{n+1}}{n}\sin(nx)$$

Example 1.3.3 (Mixed Function). Find the Fourier series of $f(x) = \begin{cases} x & \text{for } 0 \leq x \leq \pi \\ 0 & \text{for } -\pi \leq x < 0 \end{cases}$ on $[-\pi, \pi]$.

Solution: This function is neither even nor odd. We compute all coefficients:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\int_0^{\pi} x dx = \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \frac{\pi}{2}$$

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For $n \geq 1$:

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx = \frac{1}{\pi}\int_{0}^{\pi} x\cos(nx)dx$$

Using integration by parts:

$$a_n = \frac{1}{\pi}\left[\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_0^{\pi} = \frac{1}{\pi}\left(\frac{\cos(n\pi)}{n^2} - \frac{1}{n^2}\right) = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx = \frac{1}{\pi}\int_{0}^{\pi} x\sin(nx)dx$$

Using integration by parts:

$$b_n = \frac{1}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^{\pi} = \frac{1}{\pi}\left(-\frac{\pi\cos(n\pi)}{n}\right) = \frac{(-1)^{n+1}}{n}$$

Thus, the Fourier series is:

$$f(x) \sim \frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{\pi n^2}\cos(nx) + \frac{(-1)^{n+1}}{n}\sin(nx)\right]$$

1.4 Convergence of Fourier Series

So far, we have used the symbol $\sim$ to relate a function to its Fourier series. This was a deliberate choice because we have not yet established the conditions under which the infinite series actually converges, and if it does, whether it converges to the original function $f(x)$. The following sections address this crucial question.

1.4.1 Dirichlet's Conditions

The convergence of a Fourier series depends on the regularity of the function being represented. Dirichlet's conditions provide convenient sufficient hypotheses under which the Fourier series converges to a well-defined value at every point. These conditions are satisfied by many piecewise smooth periodic functions arising in mathematics, physics, and engineering.

Definition 1.4.1 (Dirichlet's Conditions)**.** Let f be a function of period $2L$, where $L > 0$. We say that f satisfies the classical Dirichlet conditions if, over one period,

1. f is absolutely integrable;
2. f has only finitely many discontinuities, each with finite one-sided limits;
3. f is monotonic on each of finitely many subintervals.

Remark 1.4.2. The third condition, together with the existence of finite one-sided limits, implies that f has bounded variation on one period. The converse need not hold: a function of bounded variation need not be piecewise monotonic with only finitely many pieces.

In particular, every piecewise continuously differentiable periodic function, with finite one-sided limits, satisfies a standard sufficient condition for Fourier convergence. The square wave, sawtooth wave, and triangular wave are familiar examples.

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1.4.2 The Riemann–Lebesgue Lemma

The Riemann–Lebesgue lemma is a fundamental result of Fourier analysis. It asserts that the integral of an absolutely integrable function against a rapidly oscillating sine or cosine tends to zero as the frequency increases.

Theorem 1.4.3 (Riemann–Lebesgue Lemma). *Let f be absolutely integrable on $[a,b]$. Then*

$$\lim_{n\to\infty}\int_a^b f(t)\cos(nt)\,dt = 0, \qquad \lim_{n\to\infty}\int_a^b f(t)\sin(nt)\,dt = 0.$$

More generally, the same conclusion holds when n is replaced by a real parameter $\lambda \to \infty$.
In particular, if f is an integrable function of period $2L$, then its Fourier coefficients satisfy

$$\lim_{n\to\infty} a_n = 0, \qquad \lim_{n\to\infty} b_n = 0.$$

Proof. We prove the assertion for the sine integral. The proof for the cosine integral is analogous.

First, suppose that ϕ is a step function on $[a,b]$, given by

$$\phi(t) = c_j, \qquad t_{j-1} < t < t_j, \quad j = 1,\dots,m,$$

where $a = t_0 < t_1 < \cdots < t_m = b$.

Then

$$\begin{aligned}\int_a^b \phi(t)\sin(nt)\,dt &= \sum_{j=1}^{m} c_j \int_{t_{j-1}}^{t_j} \sin(nt)\,dt \\ &= \frac{1}{n}\sum_{j=1}^{m} c_j\big(\cos(nt_{j-1}) - \cos(nt_j)\big).\end{aligned}$$

Consequently,

$$\left|\int_a^b \phi(t)\sin(nt)\,dt\right| \le \frac{2}{n}\sum_{j=1}^{m}|c_j|.$$

Since the sum on the right is finite and independent of n, it follows that

$$\lim_{n\to\infty}\int_a^b \phi(t)\sin(nt)\,dt = 0.$$

Now let f be any absolutely integrable function on $[a,b]$. Since step functions are dense in $L^1([a,b])$, for every $\varepsilon > 0$ there exists a step function ϕ such that

$$\int_a^b |f(t) - \phi(t)|\,dt < \frac{\varepsilon}{2}.$$

By the triangle inequality,

$$\begin{aligned}\left|\int_a^b f(t)\sin(nt)\,dt\right| &\le \left|\int_a^b (f(t)-\phi(t))\sin(nt)\,dt\right| \\ &\quad + \left|\int_a^b \phi(t)\sin(nt)\,dt\right| \\ &\le \int_a^b |f(t)-\phi(t)|\,dt + \left|\int_a^b \phi(t)\sin(nt)\,dt\right|.\end{aligned}$$

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The first term is less than $\varepsilon/2$. By the preceding result, the second term is also less than $\varepsilon/2$ for all sufficiently large n. Thus

$$\left| \int_a^b f(t) \sin(nt)\, dt \right| < \varepsilon$$

for all sufficiently large n, proving the first assertion. The cosine case follows in the same way.

The argument remains valid when n is replaced by any real parameter $\lambda \to \infty$.

Finally, the Fourier coefficients of a function of period $2L$ are

$$a_n = \frac{1}{L} \int_{-L}^{L} f(t) \cos\left(\frac{n\pi t}{L}\right) dt, \qquad b_n = \frac{1}{L} \int_{-L}^{L} f(t) \sin\left(\frac{n\pi t}{L}\right) dt.$$

Since $n\pi/L \to \infty$, the preceding result implies that $a_n \to 0$ and $b_n \to 0$. ■

1.4.3 The Sum of the Fourier Series (Convergence Theorem)

The existence of Fourier coefficients does not, by itself, guarantee convergence of the corresponding Fourier series. Under suitable regularity assumptions, however, the series converges at every point, with its sum determined by the one-sided limits of the function.

Theorem 1.4.4 (Fourier Convergence Theorem). *Let f be a function of period $2L$, where $L > 0$. Suppose that, over one period, f is piecewise continuously differentiable, with finitely many subintervals on each of which both f and f' extend continuously to the endpoints.*

Then the Fourier series of f converges at every $x \in \mathbb{R}$, and its sum is

$$\frac{f(x^+) + f(x^-)}{2},$$

where $f(x^+)$ and $f(x^-)$ denote the right-hand and left-hand limits, respectively.

In particular,

1. *at every point of continuity, the Fourier series converges to $f(x)$;*
2. *at every jump discontinuity, the Fourier series converges to the arithmetic mean of the one-sided limits.*

Proof. It is sufficient to consider a function of period 2π, since the general case follows by the change of variable $t = \pi x/L$.

Let $S_N(x)$ denote the N-th partial sum of the Fourier series of f. Thus,

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \big(a_n \cos(nx) + b_n \sin(nx)\big).$$

Substituting Euler's formulae for the coefficients and using

$$\cos(nt)\cos(nx) + \sin(nt)\sin(nx) = \cos(n(t-x)),$$

we obtain

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) D_N(t-x)\, dt,$$

where

$$D_N(u) = \frac{1}{2} + \sum_{n=1}^{N} \cos(nu) = \frac{\sin((N+\frac{1}{2})u)}{2\sin(u/2)}$$

is the Dirichlet kernel, with its value at $u = 0$ understood by continuity.

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Making the substitution $u = t - x$ and using periodicity gives

$$S_N(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+u) D_N(u)\, du.$$

Since D_N is even,

$$S_N(x) = \frac{1}{\pi} \int_{0}^{\pi} \big(f(x+u) + f(x-u)\big) D_N(u)\, du.$$

Also, by the orthogonality of the trigonometric functions,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} D_N(u)\, du = 1.$$

Equivalently,

$$\frac{2}{\pi} \int_{0}^{\pi} D_N(u)\, du = 1.$$

Set

$$A = \frac{f(x^+) + f(x^-)}{2}.$$

Subtracting A from the expression for $S_N(x)$, we obtain

$$S_N(x) - A = \frac{1}{\pi} \int_{0}^{\pi} \big(f(x+u) + f(x-u) - 2A\big) D_N(u)\, du.$$

Hence,

$$S_N(x) - A = \frac{1}{2\pi} \int_{0}^{\pi} g(u) \sin\left((N + \frac{1}{2})u\right)\, du,$$

where

$$g(u) = \frac{f(x+u) + f(x-u) - f(x^+) - f(x^-)}{\sin(u/2)}.$$

We show that g is absolutely integrable on $(0, \pi)$. By the piecewise continuous differentiability assumed for f, there exist $\delta > 0$ and $M > 0$ such that

$$|f(x+u) - f(x^+)| \le Mu, \qquad |f(x-u) - f(x^-)| \le Mu$$

for $0 < u < \delta$.

Therefore,

$$|g(u)| \le \frac{2Mu}{\sin(u/2)} \qquad (0 < u < \delta).$$

Since

$$\lim_{u \to 0^+} \frac{u}{\sin(u/2)} = 2,$$

the function g is bounded near $u = 0$.

On $[\delta, \pi]$, the denominator is bounded away from zero and the numerator is piecewise continuous. Thus $g \in L^1([0, \pi])$.

Applying the Riemann–Lebesgue lemma, with frequency $N + \frac{1}{2}$, gives

$$\lim_{N \to \infty} \int_{0}^{\pi} g(u) \sin\left((N + \frac{1}{2})u\right)\, du = 0.$$

Consequently,

$$\lim_{N \to \infty} S_N(x) = \frac{f(x^+) + f(x^-)}{2}.$$

If f is continuous at x, then $f(x^+) = f(x^-) = f(x)$, so

$$\lim_{N \to \infty} S_N(x) = f(x).$$

At a jump discontinuity, the limit is the arithmetic mean of the one-sided limits, as asserted. ■

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Example 1.4.5 (Convergence of the Square Wave)**.** Consider the 2π-periodic square wave defined by

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi. \end{cases}$$

Its Fourier series, obtained earlier, is

$$f(x) \sim \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)x)}{2n+1}.$$

The periodic extension is continuous except at $x = k\pi$, where $k \in \mathbb{Z}$. By the Fourier convergence theorem, the series converges to $f(x)$ at every point of continuity. In particular, at $x = \pi/2$, its sum is $f(\pi/2) = 1$.

At every discontinuity $x = k\pi$, the one-sided limits are -1 and 1, in some order. Hence the Fourier series converges to

$$\frac{f((k\pi)^-) + f((k\pi)^+)}{2} = \frac{-1+1}{2} = 0.$$

In particular, the series converges to zero at $x = 0$, which also follows directly from the fact that every sine term vanishes there.

Thus, the Fourier series represents the square wave at its points of continuity and assumes the midpoint value at each jump discontinuity.

Example 1.4.6 (Fourier Series and Its Convergence)**.** Let $f(x) = x$ for $-\pi < x < \pi$, extended periodically with period 2π.

1. Determine the Fourier series of f.
2. Discuss its convergence at $x = \pi/2$, $x = \pi$, and $x = -\pi$.

Solution. 1. Since f is odd, $a_0 = a_n = 0$ for all $n \geq 1$. The sine coefficients are

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx)\, dx.$$

Integrating by parts, we obtain

$$\begin{aligned} b_n &= \frac{2}{\pi} \left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi} \\ &= \frac{2(-1)^{n+1}}{n}. \end{aligned}$$

Consequently, the Fourier series is

$$f(x) \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).$$

2. The periodic extension of f is piecewise continuously differentiable. Hence the Fourier convergence theorem applies.

   (a) At $x = \pi/2$, the function is continuous. Therefore,

$$2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi}{2}\right) = \frac{\pi}{2}.$$

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(b) At $x = \pi$, the periodic extension has a jump discontinuity, with $f(\pi^-) = \pi$ and $f(\pi^+) = -\pi$. Thus, the Fourier series converges to

$$\frac{f(\pi^-) + f(\pi^+)}{2} = \frac{\pi - \pi}{2} = 0.$$

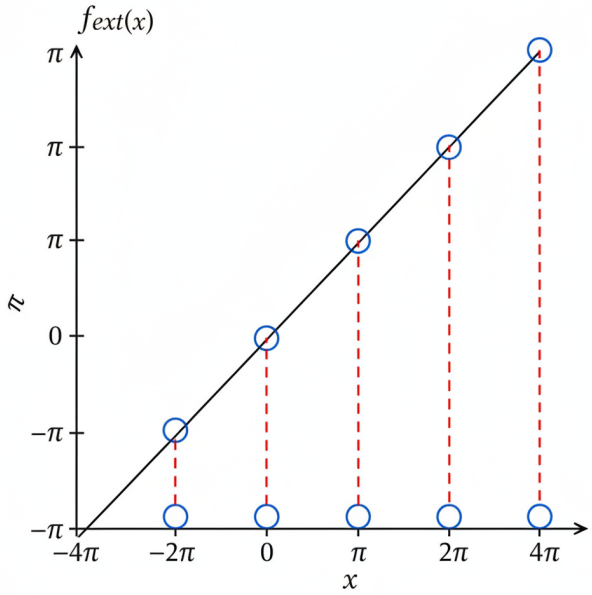
This also follows directly from $\sin(n\pi) = 0$ for every positive integer n.

(c) At $x = -\pi$, the one-sided limits of the periodic extension are $f((-\pi)^-) = \pi$ and $f((-\pi)^+) = -\pi$. Hence the Fourier series converges to

$$\frac{f((-\pi)^-) + f((-\pi)^+)}{2} = 0,$$

in agreement with $\sin(-n\pi) = 0$.

The periodic extension of $f(x) = x$ is a sawtooth wave, with jump discontinuities at the odd multiples of π, as illustrated below.



■

Example 1.4.7 (Fourier Series and Its Convergence)**.** Consider the function

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ 1, & 0 \le x < \pi, \end{cases}$$

extended periodically with period 2π.

1. Determine the Fourier series of f.
2. Discuss its convergence at $x = -\pi/2$, $x = 0$, and $x = \pi$.

Solution. 1. Since f has period 2π, its Fourier coefficients are

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx = 1.$$

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For $n \geq 1$, we have

$$a_n = \frac{1}{\pi} \int_0^{\pi} \cos(nx)\, dx = 0$$

and

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_0^{\pi} \sin(nx)\, dx \\ &= \frac{1 - \cos(n\pi)}{n\pi} = \frac{1 - (-1)^n}{n\pi}. \end{aligned}$$

Thus,

$$b_n = \begin{cases} \dfrac{2}{n\pi}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Consequently, the Fourier series of f is

$$f(x) \sim \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\sin((2k-1)x)}{2k-1}.$$

2. The periodic extension of f is piecewise continuously differentiable, with jump discontinuities at every integer multiple of π. Hence the Fourier convergence theorem applies.

   (a) At $x = -\pi/2$, the function is continuous and $f(-\pi/2) = 0$. Therefore,

   $$\frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{\sin(-(2k-1)\pi/2)}{2k-1} = 0.$$

   (b) At $x = 0$, the one-sided limits are $f(0^-) = 0$ and $f(0^+) = 1$. Hence the Fourier series converges to

   $$\frac{f(0^-) + f(0^+)}{2} = \frac{0+1}{2} = \frac{1}{2}.$$

   This also follows directly from the Fourier series, since every sine term vanishes at $x = 0$.

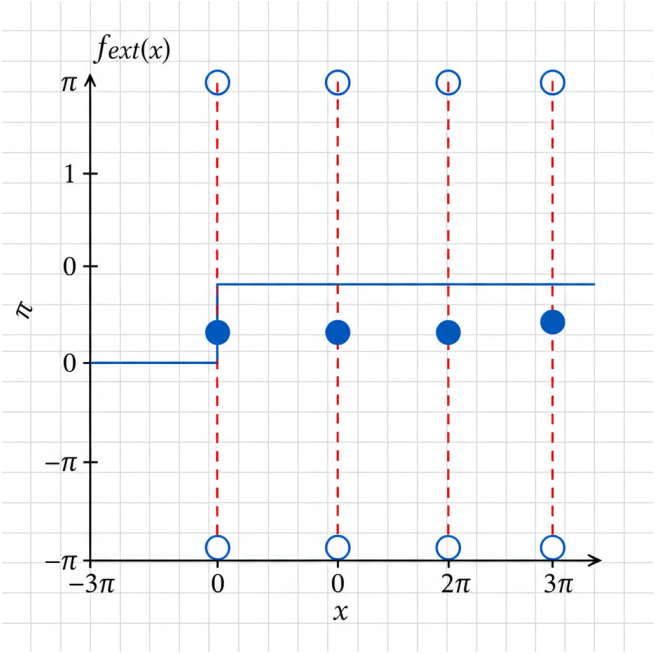
   (c) At $x = \pi$, the periodic extension has one-sided limits $f(\pi^-) = 1$ and $f(\pi^+) = 0$. Therefore, the Fourier series converges to

   $$\frac{f(\pi^-) + f(\pi^+)}{2} = \frac{1+0}{2} = \frac{1}{2},$$

   in agreement with $\sin((2k-1)\pi) = 0$ for every positive integer k.

Thus, the Fourier series converges to $f(x)$ at every point of continuity and to $1/2$ at each jump discontinuity. The function and its periodic extension are illustrated below.

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1.5 Evaluation of Infinite Series Using Fourier Expansions

A Fourier expansion may also produce identities for numerical series. Once the convergence theorem has identified the sum of the Fourier series at a chosen point, substituting a convenient value of x can eliminate many terms or replace trigonometric factors by 0, 1, or -1. This method is especially effective for sums containing reciprocals of odd integers and alternating signs. One must first verify that the Fourier series converges to the function value at the chosen point.

Fourier series expansions can be used to evaluate the sums of certain infinite series by choosing appropriate values for x in the series representation. Below are two classic examples demonstrating this technique.

Example 1.5.1. We wish to evaluate the series:

$$S_1 = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

Consider the even function $f(x) = |x|$ on the interval $[-\pi, \pi]$, extended periodically with period 2π. Its Fourier series expansion is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx),$$

where the coefficients are:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x dx = \pi, \quad a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx.$$

For $n \geq 1$, integration by parts yields:

$$a_n = \frac{2}{\pi} \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{\cos(n\pi) - 1}{n^2} = \frac{2((-1)^n - 1)}{\pi n^2}.$$

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Note that $a_n = 0$ for even n, and for odd $n = 2k-1$:

$$a_{2k-1} = \frac{2((-1)^{2k-1}-1)}{\pi(2k-1)^2} = -\frac{4}{\pi(2k-1)^2}.$$

Thus, the Fourier series becomes:

$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos((2k-1)x)}{(2k-1)^2}.$$

Setting $x = 0$ (where $f(0) = 0$):

$$0 = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{1}{(2k-1)^2}.$$

Solving for the series:

$$\sum_{k=1}^{\infty}\frac{1}{(2k-1)^2} = \frac{\pi^2}{8}.$$

Example 1.5.2. We wish to evaluate the alternating series:

$$S_2 = \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{2k-1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

Consider the odd function $f(x) = x$ on the interval $(-\pi, \pi)$, extended periodically with period 2π. Its Fourier sine series expansion is:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx),$$

where the coefficients are:

$$b_n = \frac{2}{\pi}\int_0^{\pi} x\sin(nx)dx.$$

Integration by parts yields:

$$b_n = \frac{2}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^{\pi} = \frac{2}{\pi}\cdot\left(-\frac{\pi\cos(n\pi)}{n}\right) = \frac{2(-1)^{n+1}}{n}.$$

Thus, the Fourier series becomes:

$$x = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin(nx).$$

Setting $x = \frac{\pi}{2}$ (where $f(\pi/2) = \pi/2$):

$$\frac{\pi}{2} = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{n\pi}{2}\right).$$

Note that $\sin(n\pi/2)$ is nonzero only for odd n, with:

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 1 & \text{if } n \equiv 1 \pmod 4, \\ -1 & \text{if } n \equiv 3 \pmod 4. \end{cases}$$

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Thus, the series becomes:

$$\frac{\pi}{2} = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right).$$

Solving for the series:

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

For each problem:

- Find the complete Fourier series expansion of the given function
- Choose an appropriate value of x to evaluate the series
- Justify why the Fourier series converges to the function value at your chosen point
- Manipulate the resulting equation to obtain the desired sum

Exercise 1.5.3. Use Fourier series to verify the following sums:

1. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$
2. Using the Fourier series expansion of $f(x) = x^2$ on $[-\pi, \pi]$, show that
$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$
3. Using the Fourier series expansion of $f(x) = x^4$ on $[-\pi, \pi]$, show that
$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$
4. Using the Fourier series expansion of $f(x) = |x|$ on $[-\pi, \pi]$, show that
$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

1.6 Half-Range Fourier Series Expansions

Many boundary-value problems provide a function only on a half interval. There is no unique periodic extension of such a function: the choice of extension determines the trigonometric basis and the behavior at the endpoints. Extending evenly gives a cosine series and extending oddly gives a sine series. Neither choice changes the given data at interior points of $(0, L)$, but they lead to distinct periodic functions outside that interval.

A function defined only on $[0, L]$, where $L > 0$, can be represented by a Fourier series containing exclusively cosine or sine terms. This is achieved by extending the function to $[-L, L]$ as an even or odd function and then constructing its $2L$-periodic extension.

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More precisely, the even and odd extensions of f are defined, for $0 < x < L$, by

$$f_e(x) = \begin{cases} f(-x), & -L < x < 0, \\ f(x), & 0 \le x < L, \end{cases} \qquad f_o(x) = \begin{cases} -f(-x), & -L < x < 0, \\ f(x), & 0 < x < L. \end{cases}$$

For the odd extension, we take $f_o(0) = 0$. Values at the endpoints may be assigned consistently with periodicity, without affecting the Fourier coefficients.

The Fourier series corresponding to these extensions are called **half-range Fourier expansions**. The even extension yields a cosine series, whereas the odd extension yields a sine series.

1.6.1 Fourier Cosine Series

Let f be integrable on $[0, L]$. The Fourier series of its even $2L$-periodic extension is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

where

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \qquad n \ge 0.$$

In particular, $a_0 = \frac{2}{L} \int_0^L f(x)\, dx$.

This expansion is called the **Fourier cosine series** of f. If f is piecewise continuously differentiable on $[0, L]$, the series converges to $f(x)$ at every interior point of continuity. At the endpoints, it converges to the corresponding one-sided limits of f.

1.6.2 Fourier Sine Series

The Fourier series of the odd $2L$-periodic extension of an integrable function f is

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right),$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \qquad n \ge 1.$$

This is called the **Fourier sine series** of f.

For a piecewise continuously differentiable function, the series converges to $f(x)$ at every interior point of continuity. At $x = 0$ and $x = L$, however, the sine series converges to zero, irrespective of the prescribed endpoint values of f.

> *Remark* 1.6.1 (Endpoint behavior). For a piecewise smooth function on $[0, L]$, the even extension is continuous at the endpoints whenever the corresponding one-sided limits exist and agree with the assigned values. Its cosine expansion therefore represents those endpoint limits. The odd extension changes sign on reflection: its sine series always vanishes at $x = 0$ and $x = L$. Unless $f(0) = f(L) = 0$, this creates a jump in the periodic extension and the sine series takes the midpoint value 0 there.

Example 1.6.2 (Fourier Cosine Series)**.** Find the Fourier cosine series of $f(x) = x$ on $[0, \pi]$.

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Solution. Here $L = \pi$. The constant coefficient is

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx = \pi.$$

For $n \geq 1$, integration by parts gives

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^{\pi} x\cos(nx)\,dx \\ &= \frac{2}{\pi}\left[\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_0^{\pi} \\ &= \frac{2((-1)^n - 1)}{\pi n^2}. \end{aligned}$$

Thus $a_n = 0$ for even n and $a_n = -4/(\pi n^2)$ for odd n. Consequently,

$$x = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos((2k-1)x)}{(2k-1)^2}, \qquad 0 \leq x \leq \pi.$$

The series represents the even periodic extension of f, namely $f_e(x) = |x|$ on $[-\pi, \pi]$. ■

Example 1.6.3 (Fourier Sine Series)**.** Find the Fourier sine series of $f(x) = 1$ on $[0, \pi]$.

Solution. The sine coefficients are

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi}\sin(nx)\,dx \\ &= \frac{2(1-(-1)^n)}{\pi n}. \end{aligned}$$

Hence $b_n = 0$ for even n and $b_n = 4/(\pi n)$ for odd n. Therefore,

$$1 = \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\sin((2k-1)x)}{2k-1}, \qquad 0 < x < \pi.$$

The corresponding odd periodic extension is a square wave. At $x = 0$ and $x = \pi$, the series converges to zero, which is the average of the one-sided limits of this extension. ■

Example 1.6.4 (Half-Range Expansions)**.** Find both the Fourier cosine and sine series of $f(x) = \pi - x$ on $[0, \pi]$.

Solution. **Fourier cosine series.** The constant coefficient is

$$a_0 = \frac{2}{\pi}\int_0^{\pi}(\pi - x)\,dx = \pi.$$

For $n \geq 1$, integration by parts yields

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^{\pi}(\pi - x)\cos(nx)\,dx \\ &= \frac{2}{\pi}\left[\frac{(\pi - x)\sin(nx)}{n} - \frac{\cos(nx)}{n^2}\right]_0^{\pi} \\ &= \frac{2(1-(-1)^n)}{\pi n^2}. \end{aligned}$$

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Thus,

$$\pi - x = \frac{\pi}{2} + \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos((2k-1)x)}{(2k-1)^2}, \qquad 0 \le x \le \pi.$$

Fourier sine series. For $n \ge 1$, we have

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi}(\pi - x)\sin(nx)\,dx \\ &= \frac{2}{\pi}\left[-\frac{(\pi - x)\cos(nx)}{n} - \frac{\sin(nx)}{n^2}\right]_0^{\pi} \\ &= \frac{2}{n}. \end{aligned}$$

Consequently,

$$\pi - x = 2\sum_{n=1}^{\infty}\frac{\sin(nx)}{n}, \qquad 0 < x < \pi.$$

At $x = \pi$, the sine series converges to $f(\pi) = 0$. At $x = 0$, it converges to zero, whereas $f(0) = \pi$, because the odd periodic extension has a jump discontinuity there. ■

1.6.3 Applications

Half-range Fourier expansions arise naturally in boundary value problems, particularly when the separation-of-variables method is applied to partial differential equations. Fourier sine series are associated with homogeneous Dirichlet boundary conditions, whereas Fourier cosine series arise in problems with homogeneous Neumann boundary conditions.

They are also useful in heat conduction, vibrating strings, wave propagation, and signal analysis, where symmetry and boundary conditions determine the appropriate expansion.

Exercise 1.6.5. Find the indicated half-range Fourier expansions for the following functions on $[0, \pi]$.

1. Determine the Fourier cosine series of
$$f(x) = x(\pi - x).$$
2. Determine the Fourier sine series of
$$f(x) = \begin{cases} x, & 0 \le x \le \pi/2, \\ \pi - x, & \pi/2 < x \le \pi. \end{cases}$$
3. For $f(x) = e^x$, determine
   (a) its Fourier cosine series;
   (b) its Fourier sine series.

For each problem, calculate the Fourier coefficients, obtain the corresponding series, discuss its convergence at the endpoints, and sketch the appropriate periodic extension.

1.6.4 Convergence of Fourier Coefficients and Parseval's Theorem

An important question in Fourier analysis concerns the convergence of the series formed by the squares of the Fourier coefficients. The fundamental result in this direction is Bessel's inequality, which relates the coefficients of a function to its mean-square integral.

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Theorem 1.6.6 (Bessel's Inequality). *Let f be a real-valued, square-integrable function on $[-L, L]$, where $L > 0$, with Fourier coefficients a_0, a_n, b_n. Then*

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^{L} f(x)^2 \, dx.$$

Proof. Let

$$S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

be the N-th partial sum of the Fourier series of f. Consider the mean-square error

$$E_N = \int_{-L}^{L} (f(x) - S_N(x))^2 \, dx.$$

Expanding the square, we obtain

$$E_N = \int_{-L}^{L} f(x)^2 \, dx - 2 \int_{-L}^{L} f(x) S_N(x) \, dx + \int_{-L}^{L} S_N(x)^2 \, dx.$$

By Euler's formulae and the orthogonality of the trigonometric system,

$$\int_{-L}^{L} f(x) S_N(x) \, dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^{N} (a_n^2 + b_n^2) \right).$$

Similarly, orthogonality gives

$$\int_{-L}^{L} S_N(x)^2 \, dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^{N} (a_n^2 + b_n^2) \right).$$

Consequently,

$$E_N = \int_{-L}^{L} f(x)^2 \, dx - L \left(\frac{a_0^2}{2} + \sum_{n=1}^{N} (a_n^2 + b_n^2) \right).$$

Since $E_N \geq 0$, it follows that

$$\frac{a_0^2}{2} + \sum_{n=1}^{N} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^{L} f(x)^2 \, dx.$$

Letting $N \to \infty$ yields the required inequality. ∎

Theorem 1.6.7 (Convergence of the Squares of Fourier Coefficients). *Let f be a real-valued, square-integrable function on $[-L, L]$. Then*

$$\sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

is convergent. In particular,

$$\lim_{n \to \infty} a_n = \lim_{n \to \infty} b_n = 0.$$

Proof. By Bessel's inequality,

$$0 \leq \sum_{n=1}^{N} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^{L} f(x)^2 \, dx.$$

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The partial sums form a nondecreasing sequence, since each term $a_n^2 + b_n^2$ is nonnegative. As this sequence is bounded above, it converges. Hence

$$\sum_{n=1}^{\infty} (a_n^2 + b_n^2) < \infty.$$

Therefore $a_n^2 + b_n^2 \to 0$, which implies $a_n \to 0$ and $b_n \to 0$. ■

A stronger result, known as Parseval's theorem, shows that Bessel's inequality is actually an equality for the complete trigonometric system.

Theorem 1.6.8 (Parseval's Theorem). *Let f be a real-valued, square-integrable function on $[-L, L]$, with Fourier series*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right).$$

Then

$$\frac{1}{L} \int_{-L}^{L} f(x)^2 \, dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Proof. Let S_N denote the N-th partial sum of the Fourier series of f. As established in the proof of Bessel's inequality,

$$\int_{-L}^{L} (f(x) - S_N(x))^2 \, dx = \int_{-L}^{L} f(x)^2 \, dx - L \left(\frac{a_0^2}{2} + \sum_{n=1}^{N} (a_n^2 + b_n^2) \right).$$

The trigonometric system

$$1, \quad \cos \frac{n\pi x}{L}, \quad \sin \frac{n\pi x}{L}, \qquad n \geq 1,$$

is complete in $L^2([-L, L])$. Thus, for every $\varepsilon > 0$, there exists a trigonometric polynomial

$$P(x) = \frac{c_0}{2} + \sum_{n=1}^{M} \left(c_n \cos \frac{n\pi x}{L} + d_n \sin \frac{n\pi x}{L} \right)$$

such that

$$\int_{-L}^{L} (f(x) - P(x))^2 \, dx < \varepsilon.$$

For $N \geq M$, the difference $S_N - P$ is a trigonometric polynomial of degree at most N. By the definitions of the Fourier coefficients, $f - S_N$ is orthogonal to every such polynomial. Consequently,

$$\int_{-L}^{L} (f(x) - P(x))^2 \, dx = \int_{-L}^{L} (f(x) - S_N(x))^2 \, dx + \int_{-L}^{L} (S_N(x) - P(x))^2 \, dx.$$

It follows that

$$0 \leq \int_{-L}^{L} (f(x) - S_N(x))^2 \, dx < \varepsilon$$

for every $N \geq M$. Since $\varepsilon > 0$ is arbitrary,

$$\lim_{N \to \infty} \int_{-L}^{L} (f(x) - S_N(x))^2 \, dx = 0.$$

Taking the limit in the mean-square error identity, we obtain

$$\int_{-L}^{L} f(x)^2 \, dx = L \left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right),$$

which proves Parseval's identity. ■

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Remark 1.6.9. Parseval's identity expresses the conservation of mean-square magnitude between a function and its Fourier coefficients. In particular, the Fourier partial sums converge to f in the mean-square sense, even when pointwise convergence is not guaranteed.

The completeness of the trigonometric system, used in the proof, is a standard result of Fourier analysis and can be established through the approximation of continuous periodic functions by trigonometric polynomials.

1.6.5 Worked Application of Parseval's Identity

The following example illustrates how a Fourier series can yield the value of a numerical series that is otherwise difficult to evaluate directly.

Example 1.6.10 (Evaluation of $\sum_{n=1}^{\infty} n^{-4}$)**.** Use Parseval's identity and the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$ to establish that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

Solution. Since x^2 is even, only cosine coefficients occur. As calculated earlier,

$$a_0 = \frac{2\pi^2}{3}, \qquad a_n = \frac{4(-1)^n}{n^2}, \qquad b_n = 0.$$

Parseval's identity therefore gives

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x^4\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2.$$

The integral on the left is

$$\frac{2}{\pi}\int_{0}^{\pi} x^4\,dx = \frac{2}{\pi}\left[\frac{x^5}{5}\right]_0^{\pi} = \frac{2\pi^4}{5}.$$

On the right,

$$\frac{a_0^2}{2} = \frac{2\pi^4}{9}, \qquad \sum_{n=1}^{\infty} a_n^2 = 16\sum_{n=1}^{\infty}\frac{1}{n^4}.$$

Combining these expressions,

$$16\sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{2\pi^4}{5} - \frac{2\pi^4}{9} = \frac{8\pi^4}{45},$$

and hence $\sum_{n=1}^{\infty} n^{-4} = \pi^4/90$, as required. ∎

Exercise 1.6.11. Solve the following problems related to Fourier series. Show all your work and justify each step.

1. Determine whether the following functions are even, odd, or neither:
   (a) $f(x) = x^3\cos(x)$
   (b) $g(x) = e^{-|x|}$
   (c) $h(x) = \sin(x^2)$

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2. A square wave function is defined as:

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < \pi \\ -1 & \text{for } -\pi < x < 0 \end{cases}$$

with period 2π. Find its Fourier series representation.

3. A sawtooth wave is given by:

$$f(x) = x \quad \text{for } -\pi < x < \pi$$

with period 2π. Determine its Fourier series expansion.

4. A triangular wave is defined as:

$$f(x) = \begin{cases} \pi + x & \text{for } -\pi \leq x \leq 0 \\ \pi - x & \text{for } 0 \leq x \leq \pi \end{cases}$$

with period 2π. Find its Fourier series.

5. For the function $f(x) = x^2$ on the interval $(-\pi, \pi)$:
   (a) Verify Dirichlet's conditions
   (b) Compute the Fourier coefficients a_n and b_n
   (c) Write the complete Fourier series
   (d) Using Parseval's identity, show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

6. Consider the function $f(x) = e^x$ defined on $[0, \pi]$:
   (a) Find its Fourier cosine series expansion
   (b) Find its Fourier sine series expansion
   (c) Compare the convergence rates of both series

7. For the function $f(x) = \cos(x)$ on $[0, \pi]$:
   (a) Find the half-range sine series
   (b) Find the half-range cosine series
   (c) Explain why the results differ

8. Using the Riemann-Lebesgue lemma, show that for any absolutely integrable function $f(x)$ on $[-\pi, \pi]$, the Fourier coefficients satisfy:

$$\lim_{n\to\infty} a_n = 0 \quad \text{and} \quad \lim_{n\to\infty} b_n = 0$$

9. Consider the function:

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 1 \\ 2 - x & \text{for } 1 < x \leq 2 \end{cases}$$

with period 2. Find its Fourier series representation.

10. For the function $f(x) = \sin(\pi x)$ on $[-1, 1]$:
    (a) Find its Fourier series expansion

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(b) Verify that the series converges to the function at all points

(c) Verify that all Fourier coefficients vanish except $b_1 = 1$.

11. By finding the Fourier cosine series of $f(x) = \sin x$ on $[0, \pi]$, evaluate

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{4n^2 - 1}.$$

CHAPTER

2

The Fourier Transform

The Fourier transform is a mathematical operation that transforms a function of time (or space) into a function of frequency. It decomposes a function into its constituent frequencies, similar to how a musical chord can be expressed as the frequencies of its constituent notes.

2.1 Fourier Transform: Definitions and Basic Formulas

For a function $f(t)$ that is absolutely integrable on $(-\infty, \infty)$, the Fourier transform is defined as:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt$$

where ω is the angular frequency.

Example 2.1.1 (Rectangular Pulse)**.** Find the Fourier transform of the rectangular pulse:

$$f(t) = \begin{cases} 1 & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt = \int_{-a}^{a} e^{-i\omega t}dt = \left[\frac{e^{-i\omega t}}{-i\omega}\right]_{-a}^{a} = \frac{2\sin(a\omega)}{\omega}$$

This is the well-known unnormalized sinc function: $F(\omega) = 2a \cdot \text{sinc}(a\omega)$ ■

Result 2.1.2. *The Gaussian Integral*

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Solution: Let its value be I.

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

We evaluate it by squaring the integral and converting to polar coordinates.

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$$

In polar coordinates, $x^2 + y^2 = r^2$ and the area element $dx\,dy = r\,dr\,d\theta$. The limits for the entire plane

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are $r \in [0, \infty)$ and $\theta \in [0, 2\pi]$.

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} (r\, dr)\, d\theta \\ &= \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_{r=0}^{r=\infty} d\theta \\ &= \int_0^{2\pi} \left(0 - (-\frac{1}{2}) \right) d\theta \\ &= \int_0^{2\pi} \frac{1}{2}\, d\theta = \frac{1}{2}[\theta]_0^{2\pi} = \pi \end{aligned}$$

Since $I^2 = \pi$ and the integrand e^{-x^2} is always positive, we take the positive square root.

$$I = \int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

Example 2.1.3 (Gaussian Function). Find the Fourier transform of $f(t) = e^{-at^2}$.

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-i\omega t} dt = \int_{-\infty}^{\infty} e^{-a(t^2 + \frac{i\omega}{a} t)} dt$$

Complete the square:

$$t^2 + \frac{i\omega}{a} t = \left(t + \frac{i\omega}{2a} \right)^2 + \frac{\omega^2}{4a^2}$$

Thus,

$$F(\omega) = e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} e^{-a\left(t + \frac{i\omega}{2a}\right)^2} dt = \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}$$

■

Example 2.1.4 (Exponential Decay). Find the Fourier transform of $f(t) = e^{-a|t|}$.

Solution.

$$F(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-i\omega t} dt = \int_{-\infty}^{0} e^{at} e^{-i\omega t} dt + \int_0^{\infty} e^{-at} e^{-i\omega t} dt$$

$$= \int_{-\infty}^{0} e^{(a - i\omega)t} dt + \int_0^{\infty} e^{-(a + i\omega)t} dt = \frac{1}{a - i\omega} + \frac{1}{a + i\omega} = \frac{2a}{a^2 + \omega^2}$$

■

Example 2.1.5. Consider the rectangular pulse function defined as:

$$f(t) = \begin{cases} 1 & \text{for } -\frac{1}{2} \le t \le \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $f(t)$ is given by:

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Since $f(t)$ is zero outside the interval $[-\frac{1}{2}, \frac{1}{2}]$, we can write:

$$F(\omega) = \int_{-1/2}^{1/2} e^{-i\omega t} dt$$

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Evaluating this integral:

$$F(\omega) = \left[\frac{e^{-i\omega t}}{-i\omega}\right]_{-1/2}^{1/2} = \frac{1}{-i\omega}\left(e^{-i\omega/2} - e^{i\omega/2}\right)$$

Using Euler's formula, we can express this as:

$$F(\omega) = \frac{1}{-i\omega}(-2i\sin(\omega/2)) = \frac{2}{\omega}\sin\left(\frac{\omega}{2}\right)$$

This can be written in terms of the sinc function:

$$F(\omega) = \frac{\sin(\omega/2)}{\omega/2} = \text{sinc}\left(\frac{\omega}{2}\right)$$

where the sinc function is defined as:

$$\text{sinc}(x) = \frac{\sin x}{x}$$

Thus, the Fourier transform of the rectangular pulse is:

$$\boxed{\mathcal{F}\{\text{rect}(t)\} = \text{sinc}\left(\frac{\omega}{2}\right)}$$

Definition 2.1.6 (Inverse Fourier Transform)**.** The inverse Fourier transform is given by:

$$\mathcal{F}^{-1}\{F(\omega)\} = f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}d\omega$$

2.1.1 Fourier Sine and Cosine Transforms

For real-valued functions, we often use sine and cosine transforms:

Fourier Cosine Transform:

$$\mathcal{F}_c\{f(t)\} = F_c(\omega) = \int_0^{\infty} f(t)\cos(\omega t)dt$$

Fourier Sine Transform:

$$\mathcal{F}_s\{f(t)\} = F_s(\omega) = \int_0^{\infty} f(t)\sin(\omega t)dt$$

2.2 Properties of Fourier Transform

1. **Linearity:** $\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$
2. **Time Scaling:** $\mathcal{F}\{f(at)\} = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$
3. **Time Shifting:** $\mathcal{F}\{f(t-t_0)\} = e^{-i\omega t_0}F(\omega)$
4. **Frequency Shifting:** $\mathcal{F}\{e^{i\omega_0 t}f(t)\} = F(\omega-\omega_0)$
5. **Symmetry:** If $F(\omega)$ is the Fourier transform of $f(t)$, then $\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$
6. **Derivative:** $\mathcal{F}\{f'(t)\} = i\omega F(\omega)$

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7. **Convolution Theorem:** $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$

Example 2.2.1. Let $f(t) = e^{-t^2}$ (Gaussian function). We know:

$$\mathcal{F}\{e^{-t^2}\} = \sqrt{\pi}e^{-\omega^2/4}$$

For $a = 2$, we have:

$$\mathcal{F}\{e^{-(2t)^2}\} = \mathcal{F}\{e^{-4t^2}\} = \frac{1}{2}\sqrt{\pi}e^{-(\omega/2)^2/4} = \frac{\sqrt{\pi}}{2}e^{-\omega^2/16}$$

Example 2.2.2. Let $f(t) = \text{rect}(t)$ (rectangular pulse from -1/2 to 1/2). We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

For $t_0 = 2$, we have:

$$\mathcal{F}\{\text{rect}(t-2)\} = e^{-i2\omega}\text{sinc}(\omega/2\pi)$$

The magnitude remains unchanged while the phase is shifted by -2ω.

Example 2.2.3. Let $f(t) = \text{rect}(t)$ and $\omega_0 = 4\pi$. Then:

$$\mathcal{F}\{e^{i4\pi t}\text{rect}(t)\} = \text{sinc}((\omega - 4\pi)/2\pi)$$

This represents a frequency-shifted version of the sinc function.

Example 2.2.4. Let $f(t) = e^{-|t|}$. We know:

$$\mathcal{F}\{e^{-|t|}\} = \frac{2}{1+\omega^2}$$

Then:

$$\mathcal{F}\left\{\frac{2}{1+t^2}\right\} = 2\pi e^{-|-\omega|} = 2\pi e^{-|\omega|}$$

Example 2.2.5. Let $f(t) = e^{-t}u(t)$. Then $f'(t) = -e^{-t}u(t) + \delta(t)$. We know:

$$\mathcal{F}\{e^{-t}u(t)\} = \frac{1}{1+i\omega}$$

Using the derivative property:

$$\mathcal{F}\{f'(t)\} = i\omega \cdot \frac{1}{1+i\omega} = \frac{i\omega}{1+i\omega}$$

Example 2.2.6. $\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$
Example: Let $f(t) = g(t) = \text{rect}(t)$. The convolution of two rectangular pulses is a triangular

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pulse:

$$(f * g)(t) = \begin{cases} 1 - |t| & |t| < 1 \\ 0 & |t| \geq 1 \end{cases}$$

We know:

$$\mathcal{F}\{\text{rect}(t)\} = \text{sinc}(\omega/2\pi)$$

By the convolution theorem:

$$\mathcal{F}\{f(t) * g(t)\} = [\text{sinc}(\omega/2\pi)]^2$$

Now, we prove these properties of Fourier transform one by one.

Theorem 2.2.7 (Linearity). *For any constants a and b, and functions $f(t)$ and $g(t)$ with Fourier transforms $F(\omega)$ and $G(\omega)$ respectively:*

$$\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega)$$

Proof. By the definition of the Fourier transform:

$$\mathcal{F}\{af(t) + bg(t)\} = \int_{-\infty}^{\infty} [af(t) + bg(t)]e^{-i\omega t}dt$$

Using the linearity of integration:

$$= a\int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt + b\int_{-\infty}^{\infty} g(t)e^{-i\omega t}dt = aF(\omega) + bG(\omega)$$

■

Theorem 2.2.8 (Time Scaling). *For a constant $a \neq 0$:*

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$$

Proof. Consider first the case $a > 0$. Let $u = at$, then $du = adt$, $dt = du/a$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t}dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u/a)}\frac{du}{a} = \frac{1}{a}\int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u}du = \frac{1}{a}F\left(\frac{\omega}{a}\right)$$

For $a < 0$, let $a = -|a|$. Then $u = at = -|a|t$, $du = -|a|dt$, $dt = -du/|a|$:

$$\mathcal{F}\{f(at)\} = \int_{-\infty}^{\infty} f(at)e^{-i\omega t}dt = \int_{\infty}^{-\infty} f(u)e^{-i\omega(u/a)}\left(-\frac{du}{|a|}\right) = \frac{1}{|a|}\int_{-\infty}^{\infty} f(u)e^{-i(\omega/a)u}du = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$$

Combining both cases gives the result:

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$$

■

Theorem 2.2.9 (Time Shifting). *For any real constant t_0:*

$$\mathcal{F}\{f(t - t_0)\} = e^{-i\omega t_0}F(\omega)$$

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Proof. Let $u = t - t_0$, then $t = u + t_0$, $dt = du$:

$$\mathcal{F}\{f(t-t_0)\} = \int_{-\infty}^{\infty} f(t-t_0)e^{-i\omega t}dt = \int_{-\infty}^{\infty} f(u)e^{-i\omega(u+t_0)}du$$

$$= e^{-i\omega t_0}\int_{-\infty}^{\infty} f(u)e^{-i\omega u}du = e^{-i\omega t_0}F(\omega)$$

■

Theorem 2.2.10 (Frequency Shifting). *For any real constant ω_0:*

$$\mathcal{F}\{e^{i\omega_0 t}f(t)\} = F(\omega - \omega_0)$$

Proof.

$$\mathcal{F}\{e^{i\omega_0 t}f(t)\} = \int_{-\infty}^{\infty} e^{i\omega_0 t}f(t)e^{-i\omega t}dt = \int_{-\infty}^{\infty} f(t)e^{-i(\omega-\omega_0)t}dt = F(\omega-\omega_0)$$

■

Theorem 2.2.11 (Symmetry Property of Fourier Transform). *Let $f(t)$ be a function with Fourier transform $F(\omega)$, denoted by $\mathcal{F}[f(t)] = F(\omega)$. Then the Fourier transform of the function $F(t)$ is given by:*

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

Proof. We begin with the definition of the Inverse Fourier Transform:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}\,d\omega$$

Multiplying by 2π gives:

$$2\pi f(t) = \int_{-\infty}^{\infty} F(\omega)e^{i\omega t}\,d\omega$$

To obtain the desired form, we replace the variable t with $-\omega$:

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t')e^{-it'\omega}\,dt'$$

where we have renamed the variable of integration from ω to a dummy variable t' to avoid confusion.

The integral on the right-hand side is, by definition, the Fourier transform of the function $F(t')$.

$$\int_{-\infty}^{\infty} F(t')e^{-i\omega t'}\,dt' = \mathcal{F}[F(t)]$$

Therefore, by equating the expressions, we have:

$$\mathcal{F}[F(t)] = 2\pi f(-\omega)$$

This completes the proof. ■

Theorem 2.2.12 (Symmetry). *If $F(\omega)$ is the Fourier transform of $f(t)$, then:*

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

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Proof. By definition of the Fourier transform and its inverse:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$$

Replace t with $-t$:

$$f(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} d\omega$$

Now, by definition of the Fourier transform:

$$\mathcal{F}\{F(t)\} = \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt$$

Comparing these two expressions, we see that:

$$\mathcal{F}\{F(t)\} = 2\pi f(-\omega)$$

This completes the proof. ■

Theorem 2.2.13 (Derivative). *If $f(t)$ has Fourier transform $F(\omega)$, then:*

$$\mathcal{F}\{f'(t)\} = i\omega F(\omega)$$

Proof. By definition:

$$\mathcal{F}\{f'(t)\} = \int_{-\infty}^{\infty} f'(t) e^{-i\omega t} dt$$

Using integration by parts with:

$$u = e^{-i\omega t} \qquad dv = f'(t)dt$$
$$du = -i\omega e^{-i\omega t} dt \qquad v = f(t)$$

we get:

$$\mathcal{F}\{f'(t)\} = \left[f(t) e^{-i\omega t} \right]_{-\infty}^{\infty} + i\omega \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

Since $f(t)$ must vanish at $\pm\infty$ for the Fourier transform to exist, the boundary term is zero. Thus:

$$\mathcal{F}\{f'(t)\} = i\omega \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = i\omega F(\omega)$$

This completes the proof. ■

Remark 2.2.14 (The general Case). If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}^n[f(t)] = (i\omega)^n F(\omega)$, where $f(x)$ and all its derivatives vanish at ∞.

Theorem 2.2.15 (Fubini's Theorem). *Let $h : \mathbb{R}^2 \to \mathbb{C}$ be a measurable function such that*

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |h(t,\tau)| \, d\tau \, dt < \infty.$$

Then the order of integration may be interchanged, and

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t,\tau) \, d\tau \right] dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t,\tau) \, dt \right] d\tau.$$

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Remark 2.2.16. In the proof of the convolution theorem, we apply Fubini's theorem to

$$h(t,\tau) = f(\tau)g(t-\tau)e^{-i\omega t}.$$

If $f, g \in L^1(\mathbb{R})$, then

$$\begin{aligned}\int_{\mathbb{R}}\int_{\mathbb{R}} |h(t,\tau)|\,d\tau\,dt &= \int_{\mathbb{R}} |f(\tau)| \left[\int_{\mathbb{R}} |g(t-\tau)|\,dt\right] d\tau \\ &= \|f\|_1\|g\|_1 < \infty.\end{aligned}$$

Thus, the interchange of integration is justified.

Theorem 2.2.17 (Convolution Theorem). *If $f(t)$ and $g(t)$ have Fourier transforms $F(\omega)$ and $G(\omega)$ respectively, then:*

$$\mathcal{F}\{f(t) * g(t)\} = F(\omega)G(\omega)$$

where the convolution is defined as:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

Proof. By definition:

$$\mathcal{F}\{f(t) * g(t)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau\right] e^{-i\omega t}dt$$

Interchanging the order of integration (justified by Fubini's theorem):

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(t-\tau)e^{-i\omega t}dt\right] d\tau$$

Make the substitution $u = t - \tau$ in the inner integral:

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} g(u)e^{-i\omega(u+\tau)}du\right] d\tau$$

$$= \int_{-\infty}^{\infty} f(\tau)e^{-i\omega\tau}d\tau \cdot \int_{-\infty}^{\infty} g(u)e^{-i\omega u}du = F(\omega)G(\omega)$$

This completes the proof. ■

Theorem 2.2.18 (Modulation Property). *If $F(\omega)$ is the Fourier transform of $f(t)$, then for any real constant ω_0:*

(i) $\mathcal{F}\{f(t)\cos(\omega_0 t)\} = \frac{1}{2}[F(\omega+\omega_0) + F(\omega-\omega_0)]$

(ii) $\mathcal{F}\{f(t)\sin(\omega_0 t)\} = \frac{i}{2}[F(\omega+\omega_0) - F(\omega-\omega_0)]$

Proof. (i) We begin by expressing the cosine function in terms of complex exponentials using Euler's formula:

$$\cos(\omega_0 t) = \frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2}$$

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Now, applying the Fourier transform to the modulated function:

$$\begin{aligned}\mathcal{F}\{f(t)\cos(\omega_0 t)\} &= \int_{-\infty}^{\infty} f(t)\cos(\omega_0 t)e^{-i\omega t}dt \\ &= \int_{-\infty}^{\infty} f(t)\left(\frac{e^{i\omega_0 t}+e^{-i\omega_0 t}}{2}\right)e^{-i\omega t}dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} f(t)e^{i\omega_0 t}e^{-i\omega t}dt + \frac{1}{2}\int_{-\infty}^{\infty} f(t)e^{-i\omega_0 t}e^{-i\omega t}dt \\ &= \frac{1}{2}\int_{-\infty}^{\infty} f(t)e^{-i(\omega-\omega_0)t}dt + \frac{1}{2}\int_{-\infty}^{\infty} f(t)e^{-i(\omega+\omega_0)t}dt \\ &= \frac{1}{2}F(\omega-\omega_0) + \frac{1}{2}F(\omega+\omega_0)\end{aligned}$$

■

Definition 2.2.19. The unit step function $u_0(t)$ is defined as

$$u_0(t) = \begin{cases} 0; & t < 0 \\ 1; & t > 0 \end{cases}$$

Example 2.2.20. Find the solution of the differential equation

$$y' - 2y = u_0(t)e^{-2t}, -\infty < t < \infty.$$

using the Fourier transform.

Solution. Let $\hat{y}(\omega) = \mathcal{F}\{y(t)\}(\omega)$. Using the property $\mathcal{F}\{y'(t)\}(\omega) = i\omega\hat{y}(\omega)$, we obtain:

$$(i\omega - 2)\hat{y}(\omega) = \mathcal{F}\{u_0(t)e^{-2t}\}(\omega). \tag{1}$$

Now,

$$\mathcal{F}\{u_0(t)e^{-2t}\}(\omega) = \int_0^{\infty} e^{-(2+i\omega)t}\,dt = \frac{1}{2+i\omega}.$$

Therefore,

$$(i\omega-2)\hat{y}(\omega) = \frac{1}{2+i\omega} \quad \Rightarrow \quad \hat{y}(\omega) = \frac{1}{(i\omega-2)(2+i\omega)}.$$

Note that $2 + i\omega = i\omega + 2$, so:

$$\hat{y}(\omega) = \frac{1}{(i\omega-2)(i\omega+2)} = \frac{1}{(i\omega)^2-4} = -\frac{1}{\omega^2+4}.$$

We know that,

$$\mathcal{F}\left\{e^{-2|t|}\right\}(\omega) = \frac{4}{\omega^2+4} \quad \Rightarrow \quad \mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \frac{1}{\omega^2+4}.$$

Thus,

$$\hat{y}(\omega) = -\frac{1}{\omega^2+4} = -\mathcal{F}\left\{\frac{1}{4}e^{-2|t|}\right\}(\omega) = \mathcal{F}\left\{-\frac{1}{4}e^{-2|t|}\right\}(\omega),$$

so the inverse transform gives:

$$y(t) = -\frac{1}{4}e^{-2|t|}.$$

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Verification

- For $t > 0$: $y(t) = -\frac{1}{4}e^{-2t}$, $y'(t) = \frac{1}{2}e^{-2t}$, and
$$y' - 2y = \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-2t} = e^{-2t}.$$
- For $t < 0$: $y(t) = -\frac{1}{4}e^{2t}$, $y'(t) = -\frac{1}{2}e^{2t}$, and
$$y' - 2y = -\frac{1}{2}e^{2t} + \frac{1}{2}e^{2t} = 0.$$

The solution satisfies the equation for all t.

■

Theorem 2.2.21. *Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that*
$$\int_{-\infty}^{\infty} |t^n f(t)| dt$$
converges. Then
$$\mathcal{F}[t^n f(t)](\omega) = i^n \frac{d^n}{d\omega^n} F(\omega),$$
where $F(\omega) = \mathcal{F}[f(t)](\omega)$ is the Fourier transform of $f(t)$.

Proof. We prove the result by induction on n.

For $n = 0$, the statement is trivial since $\mathcal{F}[f(t)](\omega) = F(\omega) = i^0 F^{(0)}(\omega)$.

Now, for $n = 1$, we wish to show that
$$\mathcal{F}[tf(t)](\omega) = i\frac{d}{d\omega} F(\omega).$$

Recall that
$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

Under the assumption that $\int_{-\infty}^{\infty} |tf(t)| dt$ converges, we may differentiate under the integral sign. Specifically, consider the derivative with respect to ω:
$$\frac{d}{d\omega} F(\omega) = \frac{d}{d\omega} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt.$$

Note that
$$\frac{\partial}{\partial \omega} \left(f(t) e^{-i\omega t} \right) = -itf(t) e^{-i\omega t},$$

and
$$\left| -itf(t) e^{-i\omega t} \right| = |tf(t)|.$$

Since $\int_{-\infty}^{\infty} |tf(t)| dt$ converges, the dominated convergence theorem (or Leibniz's rule for differentiation under the integral sign) applies, and we have
$$\frac{d}{d\omega} F(\omega) = \int_{-\infty}^{\infty} \frac{\partial}{\partial \omega} \left(f(t) e^{-i\omega t} \right) dt = \int_{-\infty}^{\infty} -itf(t) e^{-i\omega t} dt = -i \int_{-\infty}^{\infty} tf(t) e^{-i\omega t} dt.$$

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Therefore,

$$\int_{-\infty}^{\infty} tf(t)e^{-i\omega t}dt = i\frac{d}{d\omega}F(\omega),$$

which is exactly

$$\mathcal{F}[tf(t)](\omega) = iF'(\omega).$$

Now assume that the result holds for $n-1$, i.e.,

$$\mathcal{F}[t^{n-1}f(t)](\omega) = i^{n-1}F^{(n-1)}(\omega).$$

Define $g(t) = t^{n-1}f(t)$. Then $t^n f(t) = tg(t)$. Note that $g(t)$ is piecewise continuous on every finite interval, and by hypothesis,

$$\int_{-\infty}^{\infty} |tg(t)|dt = \int_{-\infty}^{\infty} |t^n f(t)|dt$$

converges. Therefore, we may apply the $n = 1$ case to $g(t)$ to obtain

$$\mathcal{F}[tg(t)](\omega) = i\frac{d}{d\omega}\mathcal{F}[g(t)](\omega).$$

But by the induction hypothesis,

$$\mathcal{F}[g(t)](\omega) = i^{n-1}F^{(n-1)}(\omega),$$

so

$$\mathcal{F}[t^n f(t)](\omega) = i\frac{d}{d\omega}\left(i^{n-1}F^{(n-1)}(\omega)\right) = i^n F^{(n)}(\omega).$$

This completes the induction step, and the theorem is proved. ■

Theorem 2.2.22. *Let $f(t)$ be piecewise continuous on every interval $[-l, l]$. Suppose that*

$$\int_{-\infty}^{\infty} |f(t)|\,dt$$

converges. Let $\mathcal{F}[f(t)] = F(\omega)$ and suppose that $F(0) = 0$. Then

$$\mathcal{F}\left[\int_{-\infty}^{t} f(\tau)\,d\tau\right] = \frac{1}{i\omega}F(\omega).$$

Proof. Let the function $g(t)$ be defined as the integral of $f(t)$:

$$g(t) = \int_{-\infty}^{t} f(\tau)\,d\tau$$

Our goal is to compute the Fourier transform of $g(t)$, which we denote by $G(\omega)$. By the definition of the Fourier transform, we have:

$$G(\omega) = \mathcal{F}[g(t)] = \int_{-\infty}^{\infty} g(t)e^{-i\omega t}\,dt = \int_{-\infty}^{\infty}\left(\int_{-\infty}^{t} f(\tau)\,d\tau\right)e^{-i\omega t}\,dt$$

We can solve this integral using integration by parts, where $\int u\,dv = uv - \int v\,du$. Let's choose our parts as follows:

$$u = \int_{-\infty}^{t} f(\tau)\,d\tau \qquad dv = e^{-i\omega t}\,dt$$

$$du = f(t)\,dt \qquad v = \frac{e^{-i\omega t}}{-i\omega}$$

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Applying the integration by parts formula gives:

$$G(\omega) = \left[\left(\int_{-\infty}^{t} f(\tau)\, d\tau \right) \frac{e^{-i\omega t}}{-i\omega} \right]_{t=-\infty}^{t=\infty} - \int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t)\, dt$$

Now, we must evaluate the boundary term. Let's analyze it at the limits $t \to \infty$ and $t \to -\infty$.

1. **At the upper limit ($t \to \infty$):** The term is $\lim_{t\to\infty} \left(\int_{-\infty}^{t} f(\tau)\, d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

   $$\lim_{t\to\infty} \int_{-\infty}^{t} f(\tau)\, d\tau = \int_{-\infty}^{\infty} f(\tau)\, d\tau$$

   We can relate this integral to the Fourier transform $F(\omega)$ evaluated at $\omega = 0$:

   $$F(0) = \int_{-\infty}^{\infty} f(t) e^{-i(0)t}\, dt = \int_{-\infty}^{\infty} f(t)\, dt$$

   The theorem gives the crucial condition that $F(0) = 0$. Therefore, $\int_{-\infty}^{\infty} f(\tau)\, d\tau = 0$. Since the integral part of the boundary term goes to 0 and the term $\frac{e^{-i\omega t}}{-i\omega}$ is bounded, the entire expression evaluates to 0 at the upper limit.

2. **At the lower limit ($t \to -\infty$):** The term is $\lim_{t\to-\infty} \left(\int_{-\infty}^{t} f(\tau)\, d\tau \right) \frac{e^{-i\omega t}}{-i\omega}$. The integral part becomes:

   $$\lim_{t\to-\infty} \int_{-\infty}^{t} f(\tau)\, d\tau = 0$$

   by the definition of an improper integral converging. Thus, the boundary term is also 0 at the lower limit.

Since the boundary term vanishes completely, we are left with only the integral part from our integration by parts formula:

$$\begin{aligned} G(\omega) &= -\int_{-\infty}^{\infty} \left(\frac{e^{-i\omega t}}{-i\omega} \right) f(t)\, dt \\ &= \frac{1}{i\omega} \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt \end{aligned}$$

The remaining integral is, by definition, the Fourier transform of $f(t)$, which is $F(\omega)$. Therefore, we have shown that:

$$G(\omega) = \mathcal{F}\left[\int_{-\infty}^{t} f(\tau)\, d\tau \right] = \frac{1}{i\omega} F(\omega)$$

This completes the proof. ■

Example 2.2.23 (Inverse Fourier Transform Using Convolution)**.** Using the convolution theorem, determine the inverse Fourier transform of

$$Y(\omega) = \frac{1}{12 + 7i\omega - \omega^2}.$$

Solution. Observe that

$$12 + 7i\omega - \omega^2 = (3 + i\omega)(4 + i\omega).$$

Hence

$$Y(\omega) = \frac{1}{3 + i\omega} \cdot \frac{1}{4 + i\omega}.$$

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We use the standard Fourier transform pair

$$\mathcal{F}\{e^{-at}H(t)\}(\omega) = \frac{1}{a+i\omega}, \qquad a > 0,$$

where $H(t)$ denotes the Heaviside unit step function. Thus, if $f(t) = e^{-3t}H(t)$ and $g(t) = e^{-4t}H(t)$, then $Y(\omega) = F(\omega)G(\omega)$.

By the convolution theorem,

$$y(t) = \mathcal{F}^{-1}\{Y(\omega)\} = (f * g)(t),$$

where

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,d\tau.$$

Since $f(\tau)$ vanishes for $\tau < 0$ and $g(t-\tau)$ vanishes for $\tau > t$, the convolution is zero when $t < 0$. For $t \geq 0$, we obtain

$$\begin{aligned} y(t) &= \int_0^t e^{-3\tau}e^{-4(t-\tau)}\,d\tau \\ &= e^{-4t}\int_0^t e^{\tau}\,d\tau \\ &= e^{-4t}(e^t - 1) \\ &= e^{-3t} - e^{-4t}. \end{aligned}$$

Consequently,

$$\boxed{y(t) = (e^{-3t} - e^{-4t})H(t).}$$

■

Example 2.2.24 (Differential Equation Using Fourier Transforms)**.** Solve the differential equation

$$\frac{dy}{dx} + y = e^{-|x|}, \qquad -\infty < x < \infty,$$

using the Fourier transform, assuming that y and y' are integrable on $\mathbb{R}$.

Solution. Let $Y(\omega) = \mathcal{F}\{y(x)\}(\omega)$. Taking the Fourier transform of both sides of the differential equation and using

$$\mathcal{F}\{y'\}(\omega) = i\omega Y(\omega), \qquad \mathcal{F}\{e^{-|x|}\}(\omega) = \frac{2}{1+\omega^2},$$

we obtain

$$(1 + i\omega)Y(\omega) = \frac{2}{1+\omega^2}.$$

Therefore,

$$Y(\omega) = \frac{2}{(1-i\omega)(1+i\omega)^2}.$$

To find the inverse transform, express $Y(\omega)$ in partial fractions:

$$\frac{2}{(1-i\omega)(1+i\omega)^2} = \frac{A}{1-i\omega} + \frac{B}{1+i\omega} + \frac{C}{(1+i\omega)^2}.$$

Multiplying by the denominator gives

$$2 = A(1+i\omega)^2 + B(1+\omega^2) + C(1-i\omega).$$

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Comparing coefficients of powers of $i\omega$, we find $A = B = \frac{1}{2}$ and $C = 1$. Hence

$$Y(\omega) = \frac{1}{2(1 - i\omega)} + \frac{1}{2(1 + i\omega)} + \frac{1}{(1 + i\omega)^2}.$$

We now use the standard inverse Fourier transform formulae

$$\mathcal{F}^{-1}\left\{\frac{1}{a - i\omega}\right\} = e^{ax}H(-x),$$
$$\mathcal{F}^{-1}\left\{\frac{1}{a + i\omega}\right\} = e^{-ax}H(x),$$
$$\mathcal{F}^{-1}\left\{\frac{1}{(a + i\omega)^2}\right\} = xe^{-ax}H(x),$$

where $a > 0$.

Applying these formulae with $a = 1$, we obtain

$$y(x) = \frac{1}{2}e^{x}H(-x) + \frac{1}{2}e^{-x}H(x) + xe^{-x}H(x).$$

Combining the terms on the respective half-lines, the required solution is

$$\boxed{y(x) = \begin{cases} \frac{1}{2}e^{x}, & x < 0, \\ \left(x + \frac{1}{2}\right)e^{-x}, & x \geq 0. \end{cases}}$$

To verify the result, observe that for $x < 0$,

$$y'(x) + y(x) = \frac{1}{2}e^{x} + \frac{1}{2}e^{x} = e^{x} = e^{-|x|},$$

while for $x > 0$,

$$y'(x) + y(x) = \left(\frac{1}{2} - x\right)e^{-x} + \left(x + \frac{1}{2}\right)e^{-x} = e^{-x} = e^{-|x|}.$$

Moreover, y and y' are continuous at $x = 0$, so the differential equation is satisfied throughout $\mathbb{R}$. ■

Exercise 2.2.25.

1. Find the Fourier transform of $f(t) = \delta(t - t_0)$, where δ is the Dirac delta function.
2. Find the Fourier transform of $f(t) = \cos(\omega_0 t)$.
3. Using the convolution theorem, find the Fourier transform of the triangular function:

$$f(t) = \begin{cases} 1 - |t|/a & \text{for } |t| < a \\ 0 & \text{for } |t| > a \end{cases}$$

2.3 Discrete Fourier Transform (DFT)

The Fourier transform describes a function in terms of its frequency components. In practical computations, however, signals are often available only at finitely many equally spaced points. The Discrete Fourier Transform (DFT) provides a method for analyzing the frequency content of such finite sequences.

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2.3.1 Definition of the DFT

Let $x_0, x_1, \ldots, x_{N-1}$ be a sequence of N complex numbers. Its **Discrete Fourier Transform** is the sequence $X_0, X_1, \ldots, X_{N-1}$ defined by

$$\boxed{X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N}, \qquad k = 0, 1, \ldots, N-1.}$$

Here x_n denotes the n-th sample, X_k represents the corresponding discrete frequency component, and $i^2 = -1$.

By Euler's formula, $e^{-i\theta} = \cos\theta - i\sin\theta$, the DFT can also be written as

$$X_k = \sum_{n=0}^{N-1} x_n \left(\cos\frac{2\pi k n}{N} - i \sin\frac{2\pi k n}{N} \right).$$

Thus, each coefficient X_k is obtained by combining the samples with a complex sinusoid of frequency index k.

For convenience, we introduce $\omega_N = e^{-2\pi i/N}$, called a **primitive N-th root of unity**. Since $\omega_N^N = 1$, the DFT takes the compact form

$$X_k = \sum_{n=0}^{N-1} x_n \omega_N^{kn}.$$

Remark 2.3.1. If the samples are taken at intervals of length Δt, then the sampling frequency is $f_s = 1/\Delta t$ and the DFT frequency spacing is $\Delta f = f_s/N$. The index k therefore corresponds to the frequency $k f_s/N$, interpreted modulo the sampling frequency.

2.3.2 Inverse Discrete Fourier Transform

The original samples can be recovered from their DFT coefficients by the **Inverse Discrete Fourier Transform (IDFT)**:

$$\boxed{x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i k n/N}, \qquad n = 0, 1, \ldots, N-1.}$$

The factor $1/N$ is necessary for exact reconstruction under the convention adopted here.

Theorem 2.3.2 (Inverse DFT). *If* $X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N}$, *then*

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i k n/N}.$$

Proof. Substituting the DFT formula into the proposed inverse gives

$$\begin{aligned} \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i k m/N} &= \frac{1}{N} \sum_{k=0}^{N-1} \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N} e^{2\pi i k m/N} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} x_n \sum_{k=0}^{N-1} e^{2\pi i k (m-n)/N}. \end{aligned}$$

The inner sum is N when $m = n$ and zero otherwise, by the geometric-series formula. Consequently, the expression reduces to x_m, as required. ∎

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Example 2.3.3 (DFT of a Constant Sequence)**.** Find the DFT of the sequence $x_0 = 1$, $x_1 = 1$, $x_2 = 1$, $x_3 = 1$.

Solution. Here $N = 4$, so $X_k = \sum_{n=0}^{3} e^{-2\pi ikn/4}$. For $k = 0$,

$$X_0 = 1 + 1 + 1 + 1 = 4.$$

For $k = 1$,

$$X_1 = 1 + e^{-\pi i/2} + e^{-\pi i} + e^{-3\pi i/2} = 1 - i - 1 + i = 0.$$

Similarly, $X_2 = 0$ and $X_3 = 0$. Thus the DFT is

$$\boxed{(X_0, X_1, X_2, X_3) = (4, 0, 0, 0).}$$

All the frequency content lies in the zero-frequency component because the input sequence is constant. ■

Example 2.3.4 (DFT of a Four-Point Sequence)**.** Compute the DFT of $(x_0, x_1, x_2, x_3) = (1, 2, 3, 4)$.

Solution. For $N = 4$, we have $\omega_4 = e^{-2\pi i/4} = -i$. The DFT formula becomes $X_k = \sum_{n=0}^{3} x_n \omega_4^{kn}$. Calculating the four coefficients gives

$$\begin{aligned} X_0 &= 1 + 2 + 3 + 4 = 10, \\ X_1 &= 1 + 2(-i) + 3(-1) + 4(i) = -2 + 2i, \\ X_2 &= 1 + 2(-1) + 3(1) + 4(-1) = -2, \\ X_3 &= 1 + 2(i) + 3(-1) + 4(-i) = -2 - 2i. \end{aligned}$$

Hence

$$\boxed{(X_0, X_1, X_2, X_3) = (10, -2 + 2i, -2, -2 - 2i).}$$

Notice that $X_3 = \overline{X_1}$. This illustrates the conjugate symmetry of the DFT of a real-valued sequence. ■

2.3.3 Properties of the DFT

The DFT possesses several useful properties. Throughout this subsection, let x_n and y_n have DFTs X_k and Y_k, respectively. All indices are interpreted modulo N.

1. **Linearity.** For constants $a, b \in \mathbb{C}$,

$$\mathcal{D}\{ax_n + by_n\}_k = aX_k + bY_k.$$

2. **Circular time shifting.** If $y_n = x_{(n-m) \bmod N}$, then

$$Y_k = e^{-2\pi ikm/N} X_k.$$

   Thus, a circular shift in the sample domain introduces a phase factor in the frequency domain.

3. **Frequency shifting.** If $y_n = x_n e^{2\pi imn/N}$, then

$$Y_k = X_{(k-m) \bmod N}.$$

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4. **Circular convolution.** The circular convolution of x_n and y_n is defined by

$$(x \circledast y)_n = \sum_{m=0}^{N-1} x_m y_{(n-m) \bmod N}.$$

Its DFT satisfies

$$\mathcal{D}\{x \circledast y\}_k = X_k Y_k.$$

Unlike ordinary linear convolution, circular convolution treats the sequences as periodic with period N.

5. **Parseval's identity.** The total squared magnitude is related by

$$\sum_{n=0}^{N-1} |x_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X_k|^2.$$

6. **Conjugate symmetry.** If every x_n is real, then

$$X_{(N-k) \bmod N} = \overline{X_k},$$

where the bar denotes complex conjugation. In particular, X_0 is real, and $X_{N/2}$ is also real when N is even.

Example 2.3.5 (Verification of Parseval's Identity)**.** Verify Parseval's identity for the sequence $(1, 2, 3, 4)$.

Solution. From the preceding example, its DFT is $(10, -2 + 2i, -2, -2 - 2i)$. The sum of the squares of the input samples is

$$1^2 + 2^2 + 3^2 + 4^2 = 30.$$

On the frequency side,

$$\begin{aligned} \frac{1}{4} \sum_{k=0}^{3} |X_k|^2 &= \frac{1}{4}(100 + 8 + 4 + 8) \\ &= \frac{120}{4} = 30. \end{aligned}$$

Thus, both sides are equal, verifying Parseval's identity. ■

2.4 Fast Fourier Transform (FFT)

2.4.1 Introduction

The direct calculation of an N-point DFT requires approximately N^2 arithmetic operations. For large sequences, this becomes computationally expensive.

The **Fast Fourier Transform (FFT)** is a family of efficient algorithms for calculating the DFT. A widely used FFT algorithm reduces the computational complexity from $O(N^2)$ to $O(N \log N)$.

The FFT does not define a new transform; it is a faster method of computing the same DFT coefficients.

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2.4.2 Cooley–Tukey Radix-2 FFT Algorithm

The radix-2 Cooley–Tukey algorithm applies when $N = 2^r$ for some nonnegative integer r. Its fundamental idea is to divide an N-point DFT into two DFTs of length $N/2$.

Starting with

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N},$$

we separate the even-indexed and odd-indexed samples:

$$X_k = \sum_{m=0}^{N/2-1} x_{2m} e^{-2\pi i k(2m)/N} + \sum_{m=0}^{N/2-1} x_{2m+1} e^{-2\pi i k(2m+1)/N}.$$

Define

$$E_k = \sum_{m=0}^{N/2-1} x_{2m} e^{-2\pi i k m/(N/2)}$$

and

$$O_k = \sum_{m=0}^{N/2-1} x_{2m+1} e^{-2\pi i k m/(N/2)}.$$

These are the $(N/2)$-point DFTs of the even-indexed and odd-indexed subsequences.

Writing $W_N = e^{-2\pi i/N}$, we obtain

$$X_k = E_k + W_N^k O_k, \qquad 0 \le k < N/2.$$

Moreover, since $W_N^{k+N/2} = -W_N^k$, we also have

$$X_{k+N/2} = E_k - W_N^k O_k.$$

Together, these give the fundamental **FFT combination formulae**:

$$\boxed{\begin{aligned} X_k &= E_k + W_N^k O_k, \\ X_{k+N/2} &= E_k - W_N^k O_k, \end{aligned} \qquad 0 \le k < N/2.}$$

The factors W_N^k are called **twiddle factors**.

2.4.3 Steps of the Radix-2 FFT

The radix-2 FFT can be implemented recursively as follows:

1. If $N = 1$, return the single sample.
2. Separate the sequence into its even-indexed and odd-indexed subsequences.
3. Compute the FFT of each subsequence.
4. Combine the resulting coefficients using the FFT combination formulae.

At each stage, the length of the problem is halved until sequences of length one are reached.

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Example 2.4.1 (Four-Point FFT)**.** Use the radix-2 FFT algorithm to compute the DFT of the sequence $(1, 2, 3, 4)$.

Solution. We split the sequence into its even-indexed and odd-indexed parts:

$$(x_0, x_2) = (1, 3), \qquad (x_1, x_3) = (2, 4).$$

The two-point DFT of (a, b) is $(a + b, a - b)$. Therefore, the even-indexed subsequence has DFT

$$(E_0, E_1) = (1 + 3, 1 - 3) = (4, -2),$$

while the odd-indexed subsequence has DFT

$$(O_0, O_1) = (2 + 4, 2 - 4) = (6, -2).$$

For $N = 4$, the twiddle factors are $W_4^0 = 1$ and $W_4^1 = -i$. Using the FFT combination formulae,

$$\begin{aligned} X_0 &= E_0 + O_0 = 4 + 6 = 10, \\ X_1 &= E_1 + W_4^1 O_1 = -2 + (-i)(-2) = -2 + 2i, \\ X_2 &= E_0 - O_0 = 4 - 6 = -2, \\ X_3 &= E_1 - W_4^1 O_1 = -2 - 2i. \end{aligned}$$

Thus,

$$\boxed{(X_0, X_1, X_2, X_3) = (10, -2 + 2i, -2, -2 - 2i).}$$

The result agrees exactly with the direct DFT calculation, but was obtained by combining two smaller transforms. ■

2.4.4 Computational Advantages of the FFT

The main advantage of the FFT is its reduced computational complexity. A direct DFT requires $O(N^2)$ operations, whereas a radix-2 FFT requires $O(N \log_2 N)$ operations.

For example, if $N = 1024$, the direct method involves on the order of 1024^2 arithmetic operations, while the FFT requires on the order of $1024 \log_2(1024)$ operations. This difference becomes increasingly significant as the number of samples grows.

Many FFT implementations can also operate in place, reducing storage requirements. Although the radix-2 method assumes that N is a power of two, other FFT algorithms are available for different sequence lengths. The FFT is widely used in signal processing, image analysis, numerical computation, and the efficient evaluation of convolutions.

Exercise 2.4.2.

1. Find the DFT of the sequence $(1, 0, 1, 0)$.
2. Compute the DFT of $(2, 1, 0, 1)$ directly.
3. Verify Parseval's identity for $(1, 0, 1, 0)$.
4. Use the radix-2 FFT algorithm to compute the DFT of $(1, 0, 1, 0)$.
5. Determine the circular convolution of $(1, 2, 0, 0)$ and $(1, 1, 0, 0)$.
6. Explain why the FFT is computationally preferable to the direct DFT for large N.

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CHAPTER
3

The Laplace Transform

While the Fourier series is an excellent tool for analyzing periodic functions and steady-state phenomena, many problems in engineering and physics involve functions that are not periodic. These often include transient behaviors, such as the response of a circuit immediately after a switch is flipped, or the motion of a damped spring system after an initial push.

The **Laplace Transform** is an integral transform that is particularly powerful for solving linear ordinary differential equations (ODEs), especially initial value problems. Its primary strength lies in its ability to convert a differential equation in the time domain (variable t) into an algebraic equation in a complex frequency domain (variable s). Solving the algebraic equation and then transforming the result back to the time domain is often much simpler than solving the original ODE directly.

3.1 Introduction

Definition 3.1.1 (Laplace Transform)**.** Let $f(t)$ be a function defined for all $t \geq 0$. The **Laplace transform** of $f(t)$, denoted by $\mathscr{L}\{f(t)\}$ or $F(s)$, is defined by the improper integral:

$$F(s) = \mathscr{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

provided the integral converges. The variable s is a complex variable, $s = \sigma + i\omega$, but for most introductory purposes, we can consider it a real variable. The function $F(s)$ is the transform of $f(t)$.

For the integral to exist, the function $f(t)$ must satisfy certain conditions. Specifically, $f(t)$ must be **piecewise continuous** on $[0, \infty)$ and be of **exponential order**. A function $f(t)$ is of exponential order if there exist constants $M > 0$ and c such that $|f(t)| \leq Me^{ct}$ for all $t \geq 0$. This condition essentially ensures that $f(t)$ does not grow faster than an exponential function, allowing the term e^{-st} to dominate and ensure convergence for a sufficiently large s.

3.1.1 Laplace Transforms of Elementary Functions

We derive the Laplace transforms of several basic functions directly from the definition.

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1. **The constant function,** $f(t) = 1$

$$\begin{aligned}\mathscr{L}\{1\} &= \int_0^\infty e^{-st}(1)\,dt = \left[-\frac{1}{s}e^{-st}\right]_0^\infty \\ &= \lim_{b\to\infty}\left(-\frac{1}{s}e^{-sb}\right) - \left(-\frac{1}{s}e^{0}\right) \\ &= 0 - \left(-\frac{1}{s}\right) = \frac{1}{s} \quad (\text{for } s > 0)\end{aligned}$$

2. **The exponential function,** $f(t) = e^{at}$

$$\begin{aligned}\mathscr{L}\{e^{at}\} &= \int_0^\infty e^{-st}e^{at}\,dt = \int_0^\infty e^{-(s-a)t}\,dt \\ &= \left[-\frac{1}{s-a}e^{-(s-a)t}\right]_0^\infty \\ &= 0 - \left(-\frac{1}{s-a}\right) = \frac{1}{s-a} \quad (\text{for } s > a)\end{aligned}$$

3. **The sine and cosine functions,** $f(t) = \sin(at)$ **and** $f(t) = \cos(at)$ A direct integration is possible but tedious. A more elegant method uses Euler's formula, $e^{iat} = \cos(at) + i\sin(at)$, and the known transform of the exponential function.

$$\begin{aligned}\mathscr{L}\{e^{iat}\} &= \frac{1}{s-ia} \\ &= \frac{1}{s-ia}\cdot\frac{s+ia}{s+ia} = \frac{s+ia}{s^2-(ia)^2} = \frac{s+ia}{s^2+a^2} \\ &= \frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}\end{aligned}$$

   By the linearity of the integral (and thus the transform), we also have:

$$\mathscr{L}\{e^{iat}\} = \mathscr{L}\{\cos(at) + i\sin(at)\} = \mathscr{L}\{\cos(at)\} + i\mathscr{L}\{\sin(at)\}$$

   Equating the real and imaginary parts of these two expressions gives:

$$\mathscr{L}\{\cos(at)\} = \frac{s}{s^2+a^2} \quad (\text{for } s > 0)$$

$$\mathscr{L}\{\sin(at)\} = \frac{a}{s^2+a^2} \quad (\text{for } s > 0)$$

4. **The power function,** $f(t) = t^n$ **for integer** $n \geq 0$ For $n = 1$, we use integration by parts:

$$\begin{aligned}\mathscr{L}\{t\} &= \int_0^\infty te^{-st}\,dt \quad (u = t, dv = e^{-st}dt) \\ &= \left[-\frac{t}{s}e^{-st}\right]_0^\infty - \int_0^\infty\left(-\frac{1}{s}\right)e^{-st}\,dt \\ &= (0-0) + \frac{1}{s}\int_0^\infty e^{-st}\,dt = \frac{1}{s}\mathscr{L}\{1\} = \frac{1}{s}\left(\frac{1}{s}\right) = \frac{1}{s^2} \quad (\text{for } s > 0)\end{aligned}$$

   By repeating this process, a general formula can be derived:

$$\mathscr{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad (\text{for } s > 0)$$

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3.2 Properties of the Laplace Transform

3.2.1 Linearity

The Laplace transform is a linear operator. This is its most fundamental property and follows directly from the linearity of integration.

Theorem 3.2.1 (Linearity of the Laplace Transform). *If the Laplace transforms of $f_1(t)$ and $f_2(t)$ exist, and c_1 and c_2 are any constants, then the Laplace transform of their linear combination also exists and is given by:*

$$\mathscr{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1\mathscr{L}\{f_1(t)\} + c_2\mathscr{L}\{f_2(t)\} = c_1F_1(s) + c_2F_2(s)$$

Proof. By definition,

$$\begin{aligned}
\mathscr{L}\{c_1 f_1(t) + c_2 f_2(t)\} &= \int_0^\infty e^{-st}[c_1 f_1(t) + c_2 f_2(t)]\,dt \\
&= \int_0^\infty (c_1 e^{-st} f_1(t) + c_2 e^{-st} f_2(t))\,dt \\
&= c_1\int_0^\infty e^{-st} f_1(t)\,dt + c_2\int_0^\infty e^{-st} f_2(t)\,dt \\
&= c_1\mathscr{L}\{f_1(t)\} + c_2\mathscr{L}\{f_2(t)\}
\end{aligned}$$

■

Example 3.2.2 (Using Linearity). Find the Laplace transform of $f(t) = 3 + 4\cos(2t) - 5e^{-t}$.

Solution. Using the linearity property, we can transform each term separately:

$$\begin{aligned}
\mathscr{L}\{3 + 4\cos(2t) - 5e^{-t}\} &= 3\mathscr{L}\{1\} + 4\mathscr{L}\{\cos(2t)\} - 5\mathscr{L}\{e^{-t}\} \\
&= 3\left(\frac{1}{s}\right) + 4\left(\frac{s}{s^2+2^2}\right) - 5\left(\frac{1}{s-(-1)}\right) \\
&= \frac{3}{s} + \frac{4s}{s^2+4} - \frac{5}{s+1}
\end{aligned}$$

■

Example 3.2.3 (Transforms of Hyperbolic Functions). Find the Laplace transform of $\cosh(at)$ and $\sinh(at)$.

Solution. We use the definitions $\cosh(at) = \frac{e^{at}+e^{-at}}{2}$ and $\sinh(at) = \frac{e^{at}-e^{-at}}{2}$ along with the linearity property.

$$\begin{aligned}
\mathscr{L}\{\cosh(at)\} &= \mathscr{L}\left\{\frac{1}{2}e^{at} + \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathscr{L}\{e^{at}\} + \frac{1}{2}\mathscr{L}\{e^{-at}\} \\
&= \frac{1}{2}\left(\frac{1}{s-a}\right) + \frac{1}{2}\left(\frac{1}{s+a}\right) = \frac{1}{2}\frac{(s+a)+(s-a)}{(s-a)(s+a)} \\
&= \frac{1}{2}\frac{2s}{s^2-a^2} = \frac{s}{s^2-a^2} \quad (\text{for } s > |a|)
\end{aligned}$$

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Similarly for $\sinh(at)$:

$$\begin{aligned}\mathscr{L}\{\sinh(at)\} &= \mathscr{L}\left\{\frac{1}{2}e^{at} - \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathscr{L}\{e^{at}\} - \frac{1}{2}\mathscr{L}\{e^{-at}\} \\ &= \frac{1}{2}\left(\frac{1}{s-a}\right) - \frac{1}{2}\left(\frac{1}{s+a}\right) = \frac{1}{2}\frac{(s+a)-(s-a)}{(s-a)(s+a)} \\ &= \frac{1}{2}\frac{2a}{s^2-a^2} = \frac{a}{s^2-a^2} \quad (\text{for } s > |a|)\end{aligned}$$

■

3.2.2 First Translation Theorem (s-Shifting)

This property relates the multiplication of a function by an exponential in the time domain to a shift in the frequency domain. It is extremely useful for finding transforms of functions like $e^{at}\cos(bt)$.

Theorem 3.2.4 (First Translation or s-Shifting Theorem). *If $\mathscr{L}\{f(t)\} = F(s)$, then for any real or complex number a:*

$$\mathscr{L}\{e^{at}f(t)\} = F(s-a)$$

Proof. The proof follows directly from the definition of the Laplace transform:

$$\begin{aligned}\mathscr{L}\{e^{at}f(t)\} &= \int_0^\infty e^{-st}(e^{at}f(t))\,dt \\ &= \int_0^\infty e^{-(s-a)t}f(t)\,dt\end{aligned}$$

This last integral is precisely the definition of the Laplace transform of $f(t)$, but with s replaced by $(s-a)$. Therefore,

$$\mathscr{L}\{e^{at}f(t)\} = F(s-a)$$

■

Theorem 3.2.5. *If $\mathscr{L}\{f(t)\} = F(s)$, then for any positive constant $a > 0$:*

$$\mathscr{L}\{f(t-a)u_a(t)\} = e^{-as}F(s)$$

Equivalently, for the inverse Laplace transform, if $\mathscr{L}^{-1}\{F(s)\} = f(t)$, then:

$$\mathscr{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$$

where $u_a(t)$ is the Heaviside step function:

$$u_a(t) = \begin{cases}0, & t < a \\ 1, & t \geq a\end{cases}$$

Proof. By the definition of the Laplace transform:

$$\mathscr{L}\{f(t-a)u_a(t)\} = \int_0^\infty e^{-st}f(t-a)u_a(t)dt$$

Since $u_a(t) = 0$ for $t < a$ and $u_a(t) = 1$ for $t \geq a$, the integral becomes:

$$\int_a^\infty e^{-st}f(t-a)dt$$

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Let $\tau = t - a$. Then $t = \tau + a$, and $d\tau = dt$. The limits of integration change from $t = a \to \infty$ to $\tau = 0 \to \infty$.

$$\int_0^\infty e^{-s(\tau+a)} f(\tau) d\tau$$

Factor out e^{-sa}:

$$e^{-sa} \int_0^\infty e^{-s\tau} f(\tau) d\tau$$

The integral is the definition of $F(s) = \mathscr{L}\{f(\tau)\}$.

$$e^{-sa} F(s)$$

Thus, $\mathscr{L}\{f(t-a)u_a(t)\} = e^{-as}F(s)$, which proves the theorem. ■

Example 3.2.6. Find the Laplace transform of $f(t) = e^{-3t}\sin(4t)$.

Solution. We first identify the base function, which is $f_1(t) = \sin(4t)$. Its transform is $F_1(s) = \mathscr{L}\{\sin(4t)\} = \frac{4}{s^2+16}$. Now, we apply the First Translation Theorem with $a = -3$.

$$\mathscr{L}\{e^{-3t}\sin(4t)\} = F_1(s-(-3)) = F_1(s+3)$$

We replace every s in $F_1(s)$ with $(s+3)$:

$$F(s) = \frac{4}{(s+3)^2+16} = \frac{4}{s^2+6s+9+16} = \frac{4}{s^2+6s+25}$$

■

3.2.3 Change of Scale Property

This property describes how the transform behaves when the time variable is scaled by a constant factor.

Theorem 3.2.7 (Change of Scale). *If $\mathscr{L}\{f(t)\} = F(s)$, then for any positive constant a:*

$$\mathscr{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Proof. Starting from the definition:

$$\mathscr{L}\{f(at)\} = \int_0^\infty e^{-st} f(at)\, dt$$

Let's make a substitution $\tau = at$. Then $t = \tau/a$ and $dt = d\tau/a$. The limits of integration remain unchanged.

$$\begin{aligned}\mathscr{L}\{f(at)\} &= \int_0^\infty e^{-s(\tau/a)} f(\tau)\frac{d\tau}{a} \\ &= \frac{1}{a}\int_0^\infty e^{-(s/a)\tau} f(\tau)\, d\tau\end{aligned}$$

This is the definition of the Laplace transform of $f(\tau)$ with s replaced by (s/a), multiplied by the constant $1/a$.

$$\mathscr{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

■

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3.3 Transforms of Special Functions

3.3.1 Laplace Transform of Periodic Functions

For a function that repeats itself over a fixed period, we do not need to integrate over an infinite interval. A special formula allows us to find the transform by integrating over just one period.

Theorem 3.3.1 (Transform of a Periodic Function)**.** *Let $f(t)$ be a periodic function with period $T > 0$, so that $f(t+T) = f(t)$. Then the Laplace transform of $f(t)$ is given by:*

$$\mathscr{L}\{f(t)\} = \frac{\int_0^T e^{-st} f(t)\,dt}{1 - e^{-sT}}$$

Proof. We begin by splitting the integral from the definition into a sum of integrals over each period:

$$\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = \int_0^T e^{-st} f(t)\,dt + \int_T^{2T} e^{-st} f(t)\,dt + \dots$$

Let's examine a generic term in the sum, $\int_{nT}^{(n+1)T} e^{-st} f(t)\,dt$. We use the substitution $t = u + nT$, so $dt = du$. As t goes from nT to $(n+1)T$, u goes from 0 to T.

$$\int_{nT}^{(n+1)T} e^{-st} f(t)\,dt = \int_0^T e^{-s(u+nT)} f(u+nT)\,du$$

Since $f(u+nT) = f(u)$ due to periodicity, and $e^{-s(u+nT)} = e^{-su}e^{-snT}$, this becomes:

$$e^{-snT} \int_0^T e^{-su} f(u)\,du$$

The full transform is the sum over all $n = 0, 1, 2, \dots$:

$$\begin{aligned}\mathscr{L}\{f(t)\} &= \sum_{n=0}^\infty \left(e^{-snT} \int_0^T e^{-su} f(u)\,du \right) \\ &= \left(\sum_{n=0}^\infty e^{-snT} \right) \left(\int_0^T e^{-st} f(t)\,dt \right)\end{aligned}$$

The sum is a geometric series $\sum_{n=0}^\infty (e^{-sT})^n$, which converges to $\frac{1}{1-e^{-sT}}$ for $s > 0$. Substituting this back gives the theorem's result. ∎

3.3.2 The Dirac Delta Function

The Dirac delta function, denoted $\delta(t)$, is not a function in the traditional sense but a *generalized function* or *distribution*. It is used to model phenomena of an impulsive nature, such as a point force, a point charge, or a signal of infinitesimal duration and immense amplitude.

Definition 3.3.2 (Dirac Delta Function)**.** The Dirac delta function $\delta(t-a)$ is defined by its effect under an integral, known as the **sifting property**. For any continuous function $f(t)$:

$$\int_{-\infty}^\infty f(t)\delta(t-a)\,dt = f(a)$$

The function $\delta(t-a)$ can be conceptualized as an infinitely high, infinitesimally narrow spike at

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$t = a$ with a total area of 1. That is,

$$\delta(t-a) = 0 \text{ for } t \neq a, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t-a)\, dt = 1$$

Theorem 3.3.3 (Laplace Transform of the Dirac Delta Function)**.** *For $a \geq 0$, the Laplace transform of the shifted Dirac delta function is:*

$$\mathscr{L}\{\delta(t-a)\} = e^{-as}$$

A special case is the transform of the delta function at the origin:

$$\mathscr{L}\{\delta(t)\} = 1$$

Proof. We use the definition of the Laplace transform and the sifting property of the delta function. The function being "sifted" is e^{-st}.

$$\mathscr{L}\{\delta(t-a)\} = \int_0^{\infty} e^{-st}\delta(t-a)\, dt$$

Since the delta function is zero everywhere except at $t = a$, it "samples" the function e^{-st} at the point $t = a$.

$$\mathscr{L}\{\delta(t-a)\} = e^{-sa}$$

For the case $a = 0$, we simply have $e^{-0\cdot s} = 1$. ■

Remark 3.3.4. The Dirac delta function is crucial in systems analysis. The response of a linear time-invariant system to an impulse input $\delta(t)$ is called the **impulse response** of the system. The Laplace transform of the impulse response is the system's **transfer function**, a fundamental concept in control theory and signal processing.

Laplace Transform of Derivatives

Theorem 3.3.5. *Let $f(t), t \geq 0$ be a continuous function and be of exponential order on $[0, \infty)$. Let $f'(t)$ be also of exponential order and at least piecewise continuous on $[0, \infty)$. Then*

$$\mathcal{L}[f'(t)] = s\mathcal{L}[f(t)] - f(0). \tag{3.1}$$

Proof First, consider the case when $f'(t)$ is continuous on $[0, \infty)$. Then

$$\mathcal{L}[f'(t)] = \int_0^{\infty} e^{-st} f'(t)dt = \left[e^{-st} f(t)\right]_0^{\infty} + s\int_0^{\infty} e^{-st} f(t)dt$$

$$= -f(0) + s\mathcal{L}[f(t)], s > \alpha \quad \text{(see Theorem 8.2).}$$

Hence, the result.

Suppose now, that $f'(t)$ is piecewise continuous. Assume that at $t = T$, $f'(t)$ has a finite jump. Then, we write

$$\mathcal{L}[f'(t)] = \int_0^{\infty} e^{-st} f'(t)dt = \lim_{\epsilon\to 0}\int_0^{T-\epsilon} e^{-st} f'(t)dt + \int_{T+\epsilon}^{\infty} e^{-st} f'(t)dt$$

$$= \lim_{\epsilon\to 0}\left(\left[e^{-st} f(t)\right]_0^{T-\epsilon} + s\int_0^{T-\epsilon} e^{-st} f(t)dt\right) + \lim_{\epsilon\to 0}\left(\left[e^{-st} f(t)\right]_{T+\epsilon}^{\infty} + s\int_{T+\epsilon}^{\infty} e^{-st} f(t)dt\right) \tag{3.2}$$

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It is important to note that the above theorem gives only the sufficient conditions for the existence of the Laplace transform. That is, a function may have Laplace transform even if it violates the existence conditions. For example, consider the function $f(t) = t^{-1/2}$. This function is not continuous on any interval $[0, T)$ because it has limit as $t \to 0$ from the right. However, $\int_0^T e^{-st} t^{-1/2} dt$ exists for all $T > 0$. The Laplace transform of $t^{-1/2}$ can be computed as

$$\begin{aligned}\mathcal{L}[t^{-1/2}] &= \int_0^\infty e^{-st} t^{-1/2} dt = 2\int_0^\infty e^{-u^2} du \quad (\text{setting } t = u^2)\\ &= \frac{2}{\sqrt{s}} \int_0^\infty e^{-x^2} dx \quad (\text{by setting } z = u\sqrt{s})\\ &= \sqrt{\pi/s}, \quad s > 0.\end{aligned}$$

Remark Let $f(t)$ satisfy the conditions of the existence theorem and $\mathcal{L}[f(t)] = F(s)$. Then

1. $\lim_{s\to\infty} F(s) = 0$.
2. $\lim_{s\to\infty}[sF(s)]$ is bounded.

These two results indicate that not all functions of s are Laplace transforms of some function $f(t)$.

Example 3.3.6. Find the Laplace transform of the piecewise continuous function

$$f(t) = \begin{cases} 0, & 0 \le t < 2 \\ k, & t \ge 2, \text{ k constant.} \end{cases}$$

Solution We have

$$\mathcal{L}[f(t)] = \int_0^\infty e^{-st} f(t) dt = \int_2^\infty k e^{-st} dt = k\left[-\frac{e^{-st}}{s}\right]_2^\infty = \frac{ke^{-2s}}{s}, \quad s > 0.$$

Example 3.3.7. Find the Laplace transform of $\cos^2(at)$.

Solution We have $\cos^2 at = (1 + \cos 2at)/2$. Therefore,

$$\mathcal{L}[\cos^2 at] = \mathcal{L}\left[\frac{1}{2} + \frac{1}{2}\cos 2at\right] = \frac{1}{2s} + \frac{1}{2}\frac{s}{s^2 + (2a)^2} = \frac{1}{2s} + \frac{s}{2(s^2 + 4a^2)} = \frac{s^2 + 2a^2}{s(s^2 + 4a^2)}.$$

Example 3.3.8. Find $\mathcal{L}^{-1}\left[\frac{2s+5}{s^2+2s+25}\right]$.

Solution We write $\frac{2s+5}{s^2+2s+25} = \frac{2s+5}{(s+1)^2+24}$. Then we have

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{2s+5}{(s+1)^2+24}\right] &= \mathcal{L}^{-1}\left[\frac{2(s+1)+3}{(s+1)^2+24}\right]\\ &= 2\mathcal{L}^{-1}\left[\frac{s+1}{(s+1)^2+24}\right] + \frac{3}{\sqrt{24}}\mathcal{L}^{-1}\left[\frac{\sqrt{24}}{(s+1)^2+24}\right]\\ &= 2e^{-t}\cos(\sqrt{24}t) + \frac{3}{\sqrt{24}}e^{-t}\sin(\sqrt{24}t).\end{aligned}$$

Example 3.3.9. Find $\mathcal{L}^{-1}\left[\frac{5s^2+3s-16}{(s-1)(s-2)(s+3)}\right]$.

Solution Use partial fractions to write the given function $F(s)$ as the sum of three factors.

Applications to ODE

We are now in a position to outline the procedure for solving the initial value problem

$$ay'' + by' + cy = f(t), \quad a, b, c \text{ constants}$$
$$y(0) = y_0, y'(0) = y_0'.$$

Taking the Laplace transform on both sides and denoting $\mathcal{L}[y(t)] = Y(s)$, and $\mathcal{L}[f(t)] = F(s)$, we obtain

$$a\mathcal{L}[y''] + b\mathcal{L}[y'] + c\mathcal{L}[y] = \mathcal{L}[f(t)]$$

$$a[s^2Y(s) - sy(0) - y'(0)] + b[sY(s) - y(0)] + cY(s) = F(s) \tag{3.3}$$
$$(as^2 + bs + c)Y(s) = F(s) + a(sy(0) + y'(0)) + by(0) \tag{3.4}$$
$$\text{therefore } Y(s) = \frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c} \tag{3.5}$$

which is an algebraic equation for $Y(s)$. Equation 3.5 is called the subsidiary equation. Taking the inverse Laplace transform, we obtain

$$y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c}\right] \tag{3.6}$$

Laplace transform of higher derivatives can be defined in the following way:

$$\mathcal{L}[f''(t)] = s\mathcal{L}[f'(t)] - f'(0) = s[s\mathcal{L}[f(t)] - f(0)] - f'(0) = s^2\mathcal{L}[f(t)] - sf(0) - f'(0) \tag{3.7}$$
$$\mathcal{L}[f'''(t)] = s\mathcal{L}[f''(t)] - f''(0) = s[s^2\mathcal{L}[f(t)] - sf(0) - f'(0)] - f''(0)$$
$$= s^3\mathcal{L}[f(t)] - s^2f(0) - sf'(0) - f''(0). \tag{3.8}$$

By induction, we can prove the following result.

Theorem 3.3.10. *Let $f(t), f'(t), \ldots, f^{(n-1)}(t)$ be continuous on $[0, \infty)$ and be of exponential order. Let $f^{(n)}(t)$ be at least piecewise continuous on $[0, \infty)$. Then*

$$\mathcal{L}[f^{(n)}(t)] = s^n\mathcal{L}[f(t)] - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0). \tag{3.9}$$

Proof. Left as an exercise. Use induction on n. ■

Example 3.3.11. Find the Laplace transform of $f(t) = \sin^2 t$ using the differentiation formula.

Solution We have $f(t) = \sin^2 t$ and $f(0) = 0$. Differentiating, we obtain $f'(t) = 2\sin t \cos t = \sin 2t$. Using the differentiation formula, we obtain

Laplace Transform Solution of Initial Value Problems

Laplace transformation is one of the important tools of solving the initial value problems. We first prove the following results on which the method of solution depends.

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Definition 3.3.12. A function $f(t)$ is said to be of exponential order α, if there exists a constant $M > 0$ such that $|f(t)| \leq Me^{\alpha t}$.

For example, since

$$|t| \leq e^{2t}, |e^{2t}| \leq e^{2t}, |\cos t| \leq 1, t \geq 0$$

the functions $f_1(t) = t, f_2(t) = e^{2t}$ and $f_3(t) = \cos t$ are all of exponential order. For a function of the form $f(t) = e^{ct^2}$, $c > 0$, it is not possible to determine constants α and M such that $e^{ct^2} \leq Me^{\alpha t}$. Therefore, e^{ct^2} is not a function of exponential order.

Theorem 3.3.13 (Sufficient conditions for existence of Laplace transform)**.** *If $f(t)$ is a piecewise continuous function on the interval $[0, \infty)$ and is of exponential order α for $t \geq 0$, then $\mathcal{L}[f(t)]$ exists for $s > \alpha$.*

Proof. If $f(t)$ is piecewise continuous on $[0, T]$ for $T > 0$, then $e^{-st}f(t)$ is also piecewise continuous on $[0, T]$. Hence, $\int_0^T e^{-st}f(t)dt$ exists. Therefore, the existence of the Laplace transform of $f(t)$ depends on whether this integral converges or has a finite limit as $T \to \infty$. To prove the convergence, we use the following results from the theory of improper integrals.

If for all t ($t \geq t_0$) the inequality $0 \leq \bar{f}(t) \leq g(t)$ is satisfied and if $\int_{t_0}^{\infty} g(t)dt$ converges, then $\int_{t_0}^{\infty} \bar{f}(t)dt$ also converges. If $\int_{t_0}^{\infty} |\bar{f}(t)|dt$ converges, then the integral $\int_{t_0}^{\infty} \bar{f}(t)dt$ also converges. In this case, the later integral is called an absolutely convergent integral. To use these results, we can determine a function $g(t)$ such that $\int_0^{\infty} g(t)dt$ converges and

$$|e^{-st}f(t)| \leq g(t), \quad t \geq 0.$$

Now, choose some numbers a and M such that

$$|f(t)| \leq Me^{at}.$$

Then, from Eq. (8.6), we obtain

$$|e^{-st}f(t)| \leq Me^{(a-s)t} = g(t).$$

Now,

$$\int_0^{\infty} g(t)dt = \int_0^{\infty} Me^{(a-s)t}dt$$

converges if $s > a$. Therefore, if we choose $g(t) = Me^{(a-s)t}$, $s > a$, the integral $\int_0^{\infty} e^{-st}f(t)dt$ is absolutely convergent and hence convergent. Hence, $\mathcal{L}[f(t)]$ exists. Alternatively, we have

$$|\mathcal{L}[f(t)]| = \left|\int_0^{\infty} e^{-st}f(t)dt\right| \leq \int_0^{\infty} |e^{-st}f(t)|dt \leq M\int_0^{\infty} e^{(a-s)t}dt$$

$$= M\left[\frac{e^{(a-s)t}}{a-s}\right]_0^{\infty} = \frac{M}{s-a}, \quad s > a.$$

proving the existence of the Laplace transform. ■

3.4 The Inverse Laplace Transform

The process of finding the original function $f(t)$ from its transform $F(s)$ is called the inverse Laplace transform.

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Definition 3.4.1 (Inverse Laplace Transform)**.** If $F(s)$ is the Laplace transform of a function $f(t)$, i.e., $\mathscr{L}\{f(t)\} = F(s)$, then we say that $f(t)$ is the **inverse Laplace transform** of $F(s)$, and we write:

$$f(t) = \mathscr{L}^{-1}\{F(s)\}$$

For our purposes, we find the inverse transform by recognizing the form of $F(s)$ in a table of known Laplace transform pairs and using its properties. The uniqueness of the inverse transform (for continuous functions) is guaranteed by Lerch's Theorem.

Theorem 3.4.2 (Linearity of the Inverse Transform)**.** *The inverse Laplace transform is a linear operator. For any two transforms $F_1(s)$ and $F_2(s)$ and constants c_1, c_2:*

$$\mathscr{L}^{-1}\{c_1F_1(s) + c_2F_2(s)\} = c_1\mathscr{L}^{-1}\{F_1(s)\} + c_2\mathscr{L}^{-1}\{F_2(s)\} = c_1f_1(t) + c_2f_2(t)$$

Proof. The proof follows directly from the linearity of the forward transform. Let $f_1(t) = \mathscr{L}^{-1}\{F_1(s)\}$ and $f_2(t) = \mathscr{L}^{-1}\{F_2(s)\}$. Then $\mathscr{L}\{c_1f_1(t) + c_2f_2(t)\} = c_1\mathscr{L}\{f_1(t)\} + c_2\mathscr{L}\{f_2(t)\} = c_1F_1(s) + c_2F_2(s)$. Taking the inverse transform of both sides yields the theorem. ■

Example 3.4.3 (Finding an Inverse Transform)**.** Find the inverse Laplace transform of $F(s) = \frac{5}{s-3} - \frac{2s}{s^2+4}$.

Solution. Using the linearity property, we can find the inverse of each term separately.

$$\begin{aligned} f(t) &= \mathscr{L}^{-1}\left\{\frac{5}{s-3} - \frac{2s}{s^2+4}\right\} \\ &= 5\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} - 2\mathscr{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} \end{aligned}$$

From our table of elementary transforms, we recognize these forms:

$$\mathscr{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos(at)$$

Therefore,

$$f(t) = 5e^{3t} - 2\cos(2t)$$

■

Remark 3.4.4 (Method of Partial Fractions). Often, $F(s)$ is a rational function that does not directly appear in our table. In such cases, the method of **partial fraction decomposition** is used to break down the complex fraction into a sum of simpler fractions whose inverse transforms are known. This is the most common technique for finding inverse Laplace transforms in practice.

Example 3.4.5 (Inverse Transform via Partial Fractions)**.** Find the inverse Laplace transform of $F(s) = \frac{3s+1}{s^2-s-6}$.

Solution. The denominator factors as $(s-3)(s+2)$. We decompose $F(s)$ into partial fractions:

$$\frac{3s+1}{(s-3)(s+2)} = \frac{A}{s-3} + \frac{B}{s+2}$$

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To find A, we use the cover-up method: multiply by $(s-3)$ and set $s=3$:

$$A = \left.\frac{3s+1}{s+2}\right|_{s=3} = \frac{3(3)+1}{3+2} = \frac{10}{5} = 2$$

To find B, multiply by $(s+2)$ and set $s=-2$:

$$B = \left.\frac{3s+1}{s-3}\right|_{s=-2} = \frac{3(-2)+1}{-2-3} = \frac{-5}{-5} = 1$$

So, $F(s) = \frac{2}{s-3} + \frac{1}{s+2}$. Now we can easily find the inverse transform:

$$\begin{aligned} f(t) &= \mathscr{L}^{-1}\left\{\frac{2}{s-3} + \frac{1}{s+2}\right\} \\ &= 2\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} + \mathscr{L}^{-1}\left\{\frac{1}{s-(-2)}\right\} \\ &= 2e^{3t} + e^{-2t} \end{aligned}$$

■

Example 3.4.6 (Irreducible Quadratic Factor and s-Shifting)**.** Find the inverse Laplace transform of $F(s) = \frac{s+5}{s^2+2s+5}$.

Solution. First, we inspect the denominator s^2+2s+5. The discriminant is $b^2-4ac = 2^2-4(1)(5) = 4-20 = -16 < 0$. Since the discriminant is negative, the denominator has complex roots and is an irreducible quadratic over the reals. This signals that we should complete the square and use the First Translation (s-Shifting) Theorem in reverse.

Step 1: Complete the square in the denominator.

$$s^2+2s+5 = (s^2+2s+1)+4 = (s+1)^2+4 = (s+1)^2+2^2$$

This gives

$$\begin{aligned} F(s) &= \frac{s+5}{(s+1)^2+2^2} \\ &= \frac{(s+1)+4}{(s+1)^2+2^2} \\ &= \frac{s+1}{(s+1)^2+2^2} + \frac{4}{(s+1)^2+2^2} \end{aligned}$$

We now find the inverse of each term.

- For the first term, we recognize the pattern for a shifted cosine:

$$\mathscr{L}^{-1}\left\{\frac{s+1}{(s+1)^2+2^2}\right\} \quad \text{matches the form } \mathscr{L}^{-1}\left\{\frac{s-a}{(s-a)^2+b^2}\right\} = e^{at}\cos(bt)$$

  Here, $a=-1$ and $b=2$. So, the inverse is $e^{-t}\cos(2t)$.

- For the second term, we recognize the pattern for a shifted sine:

$$\mathscr{L}^{-1}\left\{\frac{4}{(s+1)^2+2^2}\right\} = 2\cdot\mathscr{L}^{-1}\left\{\frac{2}{(s+1)^2+2^2}\right\}$$

  This matches the form $2\cdot\mathscr{L}^{-1}\left\{\frac{b}{(s-a)^2+b^2}\right\} = 2\cdot e^{at}\sin(bt)$. With $a=-1$ and $b=2$, the inverse is $2e^{-t}\sin(2t)$.

Combining the results, the final inverse transform is:

$$f(t) = \mathscr{L}^{-1}\{F(s)\} = e^{-t}\cos(2t) + 2e^{-t}\sin(2t)$$

■

Example 3.4.7 (Repeated Linear Factor)**.** Find the inverse Laplace transform of $F(s) = \frac{1}{s^2(s+1)}$.

Solution. The denominator has a repeated linear factor, s^2. The partial fraction decomposition must take the form:

$$\frac{1}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}$$

Multiplying both sides by the denominator $s^2(s+1)$ gives:

$$1 = As(s+1) + B(s+1) + Cs^2$$

We can find the constants using convenient values of s:

- Let $s = 0$: $1 = A(0) + B(0+1) + C(0) \implies B = 1$.
- Let $s = -1$: $1 = A(0) + B(0) + C(-1)^2 \implies C = 1$.
- To find A, we can expand and equate coefficients or substitute another value. Let's expand the terms with s^2: $1 = As^2 + \cdots + Cs^2$. The coefficient of s^2 on the left is 0, so $A + C = 0$. Since $C = 1$, we must have $A = -1$.

Thus, our transform is decomposed as:

$$F(s) = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1}$$

Now we find the inverse transform of each simple term using linearity:

$$\begin{aligned} f(t) &= \mathscr{L}^{-1}\left\{-\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1}\right\} \\ &= -\mathscr{L}^{-1}\left\{\frac{1}{s}\right\} + \mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} + \mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\} \end{aligned}$$

Using the standard table of transforms ($\mathscr{L}\{1\} = 1/s$, $\mathscr{L}\{t\} = 1/s^2$, $\mathscr{L}\{e^{-t}\} = 1/(s+1)$):

$$f(t) = -1 + t + e^{-t}$$

■

Example 3.4.8 (Distinct Quadratic Factors)**.** Find the inverse Laplace transform of $F(s) = \frac{s}{(s^2+1)(s^2+4)}$.

Solution. This expression can be decomposed using partial fractions. Since the factors are quadratic, the numerators must be linear terms:

$$\frac{s}{(s^2+1)(s^2+4)} = \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+4}$$

A faster method for this specific form (only even powers of s) is to let $x = s^2$ and decompose the algebraic fraction first:

$$\frac{1}{(x+1)(x+4)} = \frac{P}{x+1} + \frac{Q}{x+4}$$

Using the cover-up method:

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- $P = \frac{1}{x+4}\big|_{x=-1} = \frac{1}{3}$
- $Q = \frac{1}{x+1}\big|_{x=-4} = -\frac{1}{3}$

So, $\frac{1}{(x+1)(x+4)} = \frac{1/3}{x+1} - \frac{1/3}{x+4}$. Now substitute $x = s^2$ back and multiply by the original numerator s:

$$F(s) = s\left[\frac{1/3}{s^2+1} - \frac{1/3}{s^2+4}\right] = \frac{1}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4}$$

This form is now easily invertible.

$$\begin{aligned} f(t) &= \mathscr{L}^{-1}\left\{\frac{1}{3}\frac{s}{s^2+1^2} - \frac{1}{3}\frac{s}{s^2+2^2}\right\} \\ &= \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{s}{s^2+1^2}\right\} - \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} \end{aligned}$$

Recognizing the form for $\cos(at)$:

$$f(t) = \frac{1}{3}\cos(t) - \frac{1}{3}\cos(2t)$$

■

3.4.1 Laplace Transform of Derivatives

The single most important property for solving differential equations is the ability of the Laplace transform to convert differentiation in the time domain into multiplication by s in the frequency domain.

Theorem 3.4.9 (Transform of the First Derivative). *Let $f(t)$ be continuous and of exponential order, and let $f'(t)$ be piecewise continuous for $t \geq 0$. Then the Laplace transform of the derivative $f'(t)$ is given by:*

$$\mathscr{L}\{f'(t)\} = s\mathscr{L}\{f(t)\} - f(0) = sF(s) - f(0)$$

Proof. We use the definition of the Laplace transform and apply integration by parts.

$$\mathscr{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t)\, dt$$

Let $u = e^{-st}$ and $dv = f'(t)dt$. Then $du = -se^{-st}dt$ and $v = f(t)$. Applying the integration by parts formula, $\int u\, dv = uv - \int v\, du$:

$$\begin{aligned} \mathscr{L}\{f'(t)\} &= \left[e^{-st}f(t)\right]_0^\infty - \int_0^\infty f(t)(-se^{-st}dt) \\ &= \left(\lim_{b\to\infty} e^{-sb}f(b) - e^0 f(0)\right) + s\int_0^\infty e^{-st}f(t)\, dt \end{aligned}$$

Since $f(t)$ is of exponential order, for a sufficiently large s, the limit term $\lim_{b\to\infty} e^{-sb}f(b)$ goes to zero. The second term is $-f(0)$, and the integral is by definition $s\mathscr{L}\{f(t)\}$. Therefore,

$$\mathscr{L}\{f'(t)\} = -f(0) + sF(s) = sF(s) - f(0)$$

■

This theorem can be extended to higher-order derivatives.

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Theorem 3.4.10 (Transform of the Second Derivative). *If $f(t)$ and $f'(t)$ are continuous and $f''(t)$ is piecewise continuous, all of exponential order, then:*

$$\mathscr{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

Proof. We apply the first derivative theorem twice. Let $g(t) = f'(t)$. Then $g'(t) = f''(t)$.

$$\begin{aligned}\mathscr{L}\{f''(t)\} &= \mathscr{L}\{g'(t)\} \\ &= s\mathscr{L}\{g(t)\} - g(0) \\ &= s\mathscr{L}\{f'(t)\} - f'(0) \\ &= s(sF(s) - f(0)) - f'(0) \\ &= s^2F(s) - sf(0) - f'(0)\end{aligned}$$

■

The general formula for the n-th derivative is $\mathscr{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$.

3.4.2 Laplace Transform of Integrals

Conversely, integration in the time domain corresponds to division by s in the frequency domain.

Theorem 3.4.11 (Transform of an Integral). *If $f(t)$ is piecewise continuous and of exponential order, then:*

$$\mathscr{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}\mathscr{L}\{f(t)\} = \frac{F(s)}{s}$$

Proof. Let the integral be denoted by $g(t) = \int_0^t f(\tau)\,d\tau$. By the Fundamental Theorem of Calculus, we have $g'(t) = f(t)$. Also, it is clear from the definition that $g(0) = \int_0^0 f(\tau)\,d\tau = 0$. Now we apply the transform of the first derivative theorem to $g(t)$:

$$\mathscr{L}\{g'(t)\} = s\mathscr{L}\{g(t)\} - g(0)$$

Substituting our known values:

$$\mathscr{L}\{f(t)\} = s\mathscr{L}\left\{\int_0^t f(\tau)\,d\tau\right\} - 0$$

$$F(s) = s\mathscr{L}\left\{\int_0^t f(\tau)\,d\tau\right\}$$

Rearranging the equation for the desired term gives the result:

$$\mathscr{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{F(s)}{s}$$

■

We are now in a position to outline the procedure for solving the initial value problem using Laplace transforms:

$$\begin{gathered}ay'' + by' + cy = f(t),\ a, b, c \text{ constants} \\ y(0) = y_0, y'(0) = y_0'.\end{gathered}$$

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Taking the Laplace transform on both sides and denoting

$$\mathscr{L}[y(t)] = Y(s), \quad \text{and} \quad \mathscr{L}[f(t)] = F(s), \text{ we obtain}$$
$$a\mathscr{L}[y''] + b\mathscr{L}[y'] + c\mathscr{L}[y] = \mathscr{L}[f(t)]$$
$$\text{or} \quad a[s^2Y(s) - sy(0) - y'(0)] + b[sY(s) - y(0)] + cY(s) = F(s)$$
$$\text{or} \quad (as^2 + bs + c)Y(s) = F(s) + a(sy(0) + y'(0)) + by(0)$$
$$\text{or} \quad Y(s) = \frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c}$$

which is an algebraic equation for $Y(s)$. Taking the inverse Laplace transform, we obtain

$$y(t) = \mathscr{L}^{-1}[Y(s)] = \mathscr{L}^{-1}\left[\frac{F(s) + a(sy(0) + y'(0)) + by(0)}{as^2 + bs + c}\right]$$

Example 3.4.12. Solve the initial value problem

$$y'' + 4y = 0,\ y(0) = 1,\ y'(0) = 6.$$

Solution. Taking Laplace transform of the given differential equation, we obtain $[s^2Y(s) - sy(0) - y'(0)] + 4Y(s) = 0$, where $\mathscr{L}[y(t)] = Y(s)$. Substituting the values of $y(0)$ and $y'(0)$, we obtain

$$(s^2 + 4)Y(s) = s + 6,$$
$$\text{or } Y(s) = \frac{s+6}{s^2+4}$$
$$= \frac{s}{s^2+4} + \frac{6}{s^2+4}.$$
$$\text{Therefore, } y(t) = \mathscr{L}^{-1}[Y(s)] = \mathscr{L}^{-1}\left[\frac{s}{s^2+4}\right] + \mathscr{L}^{-1}\left[\frac{6}{s^2+4}\right] = \cos 2t + 3\sin 2t.$$

■

Example 3.4.13. Solve the initial value problem

$$y'' + 2y' - 3y = 3,\ y(0) = 4, y'(0) = -7.$$

Solution. Taking Laplace transform of the differential equation, we obtain

$$[s^2Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] - 3Y(s) = \mathscr{L}[3]$$
$$\text{or} \quad (s^2 + 2s - 3)Y(s) = \frac{3}{s} + 4s - 7 + 8 = \frac{3}{s} + 4s + 1$$
$$\text{where } \mathscr{L}[y(t)] = Y(s).$$
$$\text{Therefore, } Y(s) = \frac{3}{s(s-1)(s+3)} + \frac{4s+1}{(s-1)(s+3)}.$$

Using the partial fractions, we obtain

$$Y(s) = \left[-\frac{1}{s} + \frac{3}{4}\frac{1}{s-1} - \frac{1}{4}\frac{1}{s+3}\right] + \left[\frac{5}{4}\frac{1}{s-1} + \frac{11}{4}\frac{1}{s+3}\right]$$
$$= -\frac{1}{s} + \frac{2}{s-1} + \frac{3}{s+3}.$$

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Taking the inverse Laplace transform, we get

$$y(t) = \mathscr{L}^{-1}[Y(s)] = \mathscr{L}^{-1}\left(-\frac{1}{s}\right) + \mathscr{L}^{-1}\left(\frac{2}{s-1}\right) + \mathscr{L}^{-1}\left(\frac{3}{s+3}\right) = -1 + 2e^{t} + 3e^{-3t}.$$

■

Example 3.4.14. Solve the initial value problem $y'' + 4y' + 4y = e^{-2t}$ with $y(0) = 0$ and $y'(0) = 1$.

Solution. Let $Y(s) = \mathscr{L}\{y(t)\}$. Taking the Laplace transform of the entire equation gives:

$$\mathscr{L}\{y''\} + 4\mathscr{L}\{y'\} + 4\mathscr{L}\{y\} = \mathscr{L}\{e^{-2t}\}$$

Applying the derivative rules and substituting the initial conditions yields:

$$[s^2Y(s) - sy(0) - y'(0)] + 4[sY(s) - y(0)] + 4Y(s) = \frac{1}{s+2}$$

$$[s^2Y(s) - 1] + 4sY(s) + 4Y(s) = \frac{1}{s+2}$$

Solving for $Y(s)$ algebraically:

$$Y(s)(s^2 + 4s + 4) = 1 + \frac{1}{s+2} = \frac{s+3}{s+2}$$

$$Y(s) = \frac{s+3}{(s+2)(s+2)^2} = \frac{s+3}{(s+2)^3}$$

To find the inverse transform, we rewrite the expression as:

$$Y(s) = \frac{(s+2)+1}{(s+2)^3} = \frac{1}{(s+2)^2} + \frac{1}{(s+2)^3}$$

The inverse transform of each term is found using the s-Shifting Theorem, recalling that $\mathscr{L}^{-1}\{1/s^2\} = t$ and $\mathscr{L}^{-1}\{2/s^3\} = t^2$:

$$y(t) = e^{-2t}t + e^{-2t}\frac{t^2}{2} = te^{-2t} + \frac{1}{2}t^2e^{-2t}$$

■

Example 3.4.15. Solve the initial value problem

$$y'' - 5y' + 4y = e^{2t},\ y(0) = 19/12, y'(0) = 8/3.$$

Theorem 3.4.16. *Let $f(t)$ be piecewise continuous on $[0,\infty)$ and be of exponential order. Further, let $\lim_{t\to 0^+}[f(t)/t]$ exist. Then*

$$\mathscr{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)ds, \quad s > \alpha$$

where $\mathscr{L}[f(t)] = F(s)$.

Proof. From the definition, we have

$$F(s) = \int_0^\infty e^{-st}f(t)dt.$$

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Integrating from s to ∞, we obtain

$$\int_s^\infty F(s)ds = \int_s^\infty \left[\int_0^\infty e^{-st} f(t)dt\right] ds.$$

Now, $f(t)$ is piecewise continuous, is of exponential order and limit of $f(t)/t$ exists as t approaches zero from the right. Under these assumptions, the order of integration can be reversed and we can write

$$\begin{aligned}\int_s^\infty F(s)ds &= \int_0^\infty \left[\int_s^\infty e^{-st}ds\right] f(t)dt \\ &= \int_0^\infty \left[\frac{e^{-st}}{-t}\right]_s^\infty f(t)dt \\ &= \int_0^\infty \left[e^{-st}\frac{f(t)}{t}\right] dt = \mathscr{L}\left[\frac{f(t)}{t}\right], \quad s > \alpha.\end{aligned}$$

■

Example 3.4.17. Find the Laplace transform of $(\sin \omega t)/t$.

Solution Using above Theorem, we obtain

$$\begin{aligned}\mathscr{L}\left[\frac{\sin \omega t}{t}\right] &= \int_s^\infty \mathscr{L}[\sin \omega t]ds \\ &= \int_s^\infty \frac{\omega}{s^2+\omega^2}ds \\ &= \left[\tan^{-1}\left(\frac{s}{\omega}\right)\right]_s^\infty \\ &= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{\omega}\right) \\ &= \cot^{-1}\left(\frac{s}{\omega}\right).\end{aligned}$$

Definition 3.4.18 (Convolution Integral)**.** The convolution of two functions $f(t)$ and $g(t)$, denoted by $(f * g)(t)$, is defined as:

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$$

Theorem 3.4.19 (Convolution Theorem)**.** *If $\mathscr{L}\{f(t)\} = F(s)$ and $\mathscr{L}\{g(t)\} = G(s)$, then the Laplace transform of their convolution is the product of their transforms:*

$$\mathscr{L}\{(f * g)(t)\} = F(s)G(s)$$

In its inverse form, which is more commonly used for problem-solving:

$$\mathscr{L}^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$$

Proof. By the definition of the Laplace transform, we have:

$$\mathscr{L}\{(f * g)(t)\} = \int_0^\infty e^{-st}(f * g)(t)\,dt$$

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Substituting the definition of the convolution integral, $(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$:

$$\mathscr{L}\{(f*g)(t)\} = \int_0^\infty e^{-st}\left[\int_0^t f(\tau)g(t-\tau)\,d\tau\right]dt$$

This is a double integral over the region in the t-τ plane defined by $0 \le \tau \le t < \infty$. We can change the order of integration. The new limits will be $\tau \le t < \infty$ for the inner integral and $0 \le \tau < \infty$ for the outer integral.

$$\begin{aligned}\mathscr{L}\{(f*g)(t)\} &= \int_0^\infty \int_\tau^\infty e^{-st} f(\tau)g(t-\tau)\,dt\,d\tau \\ &= \int_0^\infty f(\tau)\left[\int_\tau^\infty e^{-st}g(t-\tau)\,dt\right]d\tau\end{aligned}$$

Let's evaluate the inner integral by making the substitution $u = t - \tau$. This implies $t = u + \tau$ and $dt = du$. The limits of integration for u become 0 to ∞.

$$\begin{aligned}\int_\tau^\infty e^{-st}g(t-\tau)\,dt &= \int_0^\infty e^{-s(u+\tau)}g(u)\,du \\ &= \int_0^\infty e^{-su}e^{-s\tau}g(u)\,du \\ &= e^{-s\tau}\int_0^\infty e^{-su}g(u)\,du \\ &= e^{-s\tau}G(s)\end{aligned}$$

Substituting this result back into the outer integral:

$$\begin{aligned}\mathscr{L}\{(f*g)(t)\} &= \int_0^\infty f(\tau)[e^{-s\tau}G(s)]\,d\tau \\ &= G(s)\int_0^\infty e^{-s\tau}f(\tau)\,d\tau\end{aligned}$$

The remaining integral is, by definition, the Laplace transform of $f(t)$, which is $F(s)$. Therefore,

$$\mathscr{L}\{(f*g)(t)\} = G(s)F(s) = F(s)G(s)$$

This completes the proof. ■

Example 3.4.20. Find the inverse Laplace transform of $H(s) = \frac{1}{s(s^2+4)}$ using convolution.

Solution. We express $H(s)$ as the product of $F(s) = \frac{1}{s}$ and $G(s) = \frac{1}{s^2+4}$. The corresponding inverse transforms are $f(t) = 1$ and $g(t) = \frac{1}{2}\sin(2t)$. Applying the convolution theorem, the inverse transform $h(t)$ is given by the integral:

$$\begin{aligned}h(t) = (g*f)(t) &= \int_0^t g(\tau)f(t-\tau)\,d\tau = \int_0^t \frac{1}{2}\sin(2\tau)\cdot(1)\,d\tau \\ &= \frac{1}{2}\left[-\frac{\cos(2\tau)}{2}\right]_0^t = -\frac{1}{4}(\cos(2t)-\cos(0)) \\ &= \frac{1}{4}(1-\cos(2t))\end{aligned}$$

■

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Theorem 3.4.21. *Using convolution, solve the initial value problem*

$$y'' + 9y = \sin 3t, \quad y(0) = 0, \quad y'(0) = 0.$$

Solution. Taking Laplace transform of the differential equation, we obtain

$$s^2Y(s) + 9Y(s) = \frac{3}{s^2+9}, \quad \text{or} \quad Y(s) = \frac{3}{(s^2+9)^2}.$$

Write $Y(s)$ as

$$Y(s) = \frac{1}{3}\left(\frac{3}{s^2+9}\right)\left(\frac{3}{s^2+9}\right) = \frac{1}{3}F(s)G(s)$$

where $F(s) = G(s) = 3/(s^2+9)$. We have $f(t) = g(t) = \sin 3t$. Using the convolution theorem, we obtain

$$\begin{aligned}
y(t) &= \mathscr{L}^{-1}\left[\frac{1}{3}F(s)G(s)\right] = \frac{1}{3}\int_0^t \sin 3\tau \sin 3(t-\tau)d\tau \\
&= \frac{1}{6}\int_0^t [\cos(3\tau - 3(t-\tau)) - \cos(3\tau + 3(t-\tau))]d\tau \\
&= \frac{1}{6}\int_0^t [\cos(6\tau - 3t) - \cos 3t]d\tau \\
&= \frac{1}{6}\left[\frac{\sin(6\tau - 3t)}{6} - \tau\cos 3t\right]_0^t \\
&= \frac{1}{6}\left[\frac{\sin(6t-3t)}{6} - t\cos 3t - \left(\frac{\sin(-3t)}{6} - 0\right)\right] \\
&= \frac{1}{6}\left[\frac{\sin 3t}{6} - t\cos 3t + \frac{\sin 3t}{6}\right] \\
&= \frac{1}{6}\left[\frac{2\sin 3t}{6} - t\cos 3t\right] \\
&= \frac{1}{6}\left[\frac{1}{3}\sin 3t - t\cos 3t\right] \\
&= \frac{1}{18}\sin 3t - \frac{1}{6}t\cos 3t.
\end{aligned}$$

■

3.5 Applications to Partial Differential Equations

The Laplace transform can simplify linear partial differential equations by transforming one variable (typically time t), thereby reducing the PDE to an ODE in the remaining spatial variable(s). The key relations for a function $u(x,t)$ with transform $U(x,s) = \mathscr{L}\{u(x,t)\}$ are:

$$\mathscr{L}\left\{\frac{\partial u}{\partial t}\right\} = sU(x,s) - u(x,0) \quad \text{and} \quad \mathscr{L}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \frac{d^2U}{dx^2}$$

Example 3.5.1. Using Laplace transforms, find the solution of the initial value problem

$$x\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = xt, \quad u(x,0) = 0, u(0,t) = t.$$

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Solution. Denote $\mathscr{L}[u(x,t)] = U(x,s)$. Taking Laplace transforms (with respect to t) of the differential equation and the boundary condition $u(0,t) = t$, we obtain

$$sxU(x,s) - u(x,0)] + \frac{d}{dx}U(x,s) = \frac{x}{s^2}. \quad U(0,s) = \frac{1}{s^2}$$

or

$$\frac{dU}{dx} + sxU = \frac{x}{s^2}.$$

The integrating factor of this ordinary differential equation is $e^{sx^2/2}$. Integrating, we obtain

$$e^{sx^2/2}U = \int \frac{x}{s^2}e^{sx^2/2}dx + a(s) = \frac{1}{s^3}e^{sx^2/2} + a(s)$$

or

$$U(x,s) = \frac{1}{s^3} + a(s)e^{-sx^2/2}$$

where $a(s)$ is an arbitrary function of s. Using the condition $U(0,s) = 1/s^2$, we obtain

$$U(0,s) = \frac{1}{s^3} + a(s), \quad \text{or} \quad a(s) = \frac{1}{s^2} - \frac{1}{s^3}.$$

Hence,

$$U(x,s) = \frac{1}{s^3} + \left(\frac{1}{s^2} - \frac{1}{s^3}\right)e^{-sx^2/2}.$$

Therefore,

$$u(x,t) = \frac{t^2}{2!} + \mathscr{L}^{-1}\left[\left(\frac{1}{s^2} - \frac{1}{s^3}\right)e^{-sx^2/2}\right].$$

Using the 2nd shift theorem [Theorem 3.2.5], we obtain

$$u(x,t) = \frac{t^2}{2!} + \left[\left(t - \frac{x^2}{2}\right) - \frac{1}{2!}\left(t - \frac{x^2}{2}\right)^2\right]u_{x^2/2}(t)$$

$$= \begin{cases} \frac{t^2}{2}, & t < \frac{x^2}{2} \\ \frac{t^2}{2} + \left(t - \frac{x^2}{2}\right) - \frac{1}{2!}\left(t - \frac{x^2}{2}\right)^2, & t \geq \frac{x^2}{2}. \end{cases}$$

■

Example 3.5.2 (Application to the Heat Equation)**.** Solve the heat conduction problem in a semi-infinite rod ($x \geq 0$) governed by:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad \text{with } u(x,0) = 0,\ u(0,t) = u_0,\ \lim_{x\to\infty} u(x,t) = 0$$

Solution. Applying the Laplace transform with respect to t to the PDE and its boundary conditions converts the problem into an ordinary differential equation for $U(x,s)$. The PDE becomes:

$$sU(x,s) - u(x,0) = k\frac{d^2U}{dx^2} \implies \frac{d^2U}{dx^2} - \frac{s}{k}U = 0$$

The transformed boundary conditions are $U(0,s) = u_0/s$ and $\lim_{x\to\infty} U(x,s) = 0$. The general solution to this ODE is $U(x,s) = C_1e^{\sqrt{s/k}x} + C_2e^{-\sqrt{s/k}x}$. The boundary condition at infinity requires $C_1 = 0$ to ensure a bounded solution. The condition at $x = 0$ then gives $C_2 = u_0/s$. The solution in the s-domain is therefore:

$$U(x,s) = \frac{u_0}{s}e^{-x\sqrt{s/k}}$$

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The inverse transform of this expression is non-trivial and can be found in standard tables:

$$u(x,t) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right)$$

where erfc is the complementary error function. The method successfully reduces the PDE to a solvable ODE. ■

Example 3.5.3. Find the inverse Laplace transform of $H(s) = \frac{s}{(s^2+a^2)^2}$ using the Convolution Theorem.

Solution. We can express $H(s)$ as the product of $F(s) = \frac{s}{s^2+a^2}$ and $G(s) = \frac{1}{s^2+a^2}$. The inverse transforms of these functions are:

$$f(t) = \mathscr{L}^{-1}\{F(s)\} = \cos(at)$$

$$g(t) = \mathscr{L}^{-1}\{G(s)\} = \frac{1}{a}\sin(at)$$

By the Convolution Theorem, $h(t) = \mathscr{L}^{-1}\{H(s)\}$ is given by the convolution integral $(f * g)(t)$:

$$h(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau = \int_0^t \cos(a\tau)\left(\frac{1}{a}\sin(a(t-\tau))\right)\,d\tau$$

Using the product-to-sum trigonometric identity $\sin(A)\cos(B) = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$, where $A = a(t-\tau)$ and $B = a\tau$:

$$\begin{aligned} h(t) &= \frac{1}{a}\int_0^t \frac{1}{2}[\sin(at - a\tau + a\tau) + \sin(at - a\tau - a\tau)]\,d\tau \\ &= \frac{1}{2a}\int_0^t [\sin(at) + \sin(at - 2a\tau)]\,d\tau \end{aligned}$$

We integrate with respect to τ, treating t as a constant:

$$\begin{aligned} h(t) &= \frac{1}{2a}\left[\tau\sin(at) + \frac{\cos(at-2a\tau)}{2a}\right]_0^t \\ &= \frac{1}{2a}\left[\left(t\sin(at) + \frac{\cos(-at)}{2a}\right) - \left(0 + \frac{\cos(at)}{2a}\right)\right] \\ &= \frac{1}{2a}\left[t\sin(at) + \frac{\cos(at)}{2a} - \frac{\cos(at)}{2a}\right] \\ &= \frac{t}{2a}\sin(at) \end{aligned}$$

This result is a standard entry in Laplace transform tables and is fundamental in the study of resonance. ■

Example 3.5.4. Solve the initial value problem $y'' + y = \cos(2t)$ with initial conditions $y(0) = 1$ and $y'(0) = 0$.

Solution. Let $Y(s) = \mathscr{L}\{y(t)\}$. Taking the Laplace transform of the equation and applying the initial conditions gives:

$$[s^2Y(s) - sy(0) - y'(0)] + Y(s) = \mathscr{L}\{\cos(2t)\}$$

$$s^2Y(s) - s(1) - 0 + Y(s) = \frac{s}{s^2+4}$$

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Now, we solve algebraically for $Y(s)$:

$$Y(s)(s^2+1) = s + \frac{s}{s^2+4}$$

$$Y(s) = \frac{s}{s^2+1} + \frac{s}{(s^2+1)(s^2+4)}$$

We use partial fractions on the second term. As shown in a previous example, this decomposition yields:

$$\frac{s}{(s^2+1)(s^2+4)} = \frac{1}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4}$$

Substituting this back into the expression for $Y(s)$:

$$Y(s) = \frac{s}{s^2+1} + \left(\frac{1}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4}\right) = \frac{4}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4}$$

Finally, we find the inverse Laplace transform term-by-term:

$$\begin{aligned} y(t) &= \mathscr{L}^{-1}\left\{\frac{4}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4}\right\} \\ &= \frac{4}{3}\mathscr{L}^{-1}\left\{\frac{s}{s^2+1^2}\right\} - \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} \\ &= \frac{4}{3}\cos(t) - \frac{1}{3}\cos(2t) \end{aligned}$$

The solution consists of the natural frequency oscillation ($\cos t$) and an oscillation at the driving frequency ($\cos 2t$). ■

Example 3.5.5 (Application to the Wave Equation)**.** A semi-infinite elastic string ($x \geq 0$) is initially at rest and in its equilibrium position. At time $t = 0$, the end at $x = 0$ is forced to undergo sinusoidal motion given by $f(t) = \sin(\omega t)$. Find the resulting displacement $u(x,t)$.

Solution. The problem is described by the one-dimensional wave equation and its associated conditions:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Conditions: $u(x,0) = 0$, $u_t(x,0) = 0$, $u(0,t) = \sin(\omega t)$, and $\lim_{x\to\infty} u(x,t)$ is bounded.

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. We transform the PDE with respect to t:

$$s^2 U(x,s) - s u(x,0) - u_t(x,0) = c^2 \frac{d^2U}{dx^2}$$

Substituting the initial conditions simplifies this to an ODE in x:

$$s^2 U(x,s) = c^2 \frac{d^2U}{dx^2} \implies \frac{d^2U}{dx^2} - \left(\frac{s}{c}\right)^2 U = 0$$

The transformed boundary conditions are $U(0,s) = \mathscr{L}\{\sin(\omega t)\} = \frac{\omega}{s^2+\omega^2}$ and $\lim_{x\to\infty} U(x,s) = 0$. The general solution to the ODE is $U(x,s) = C_1 e^{(s/c)x} + C_2 e^{-(s/c)x}$. The boundedness condition as $x \to \infty$ forces $C_1 = 0$. Applying the condition at $x = 0$ gives $C_2 = U(0,s) = \frac{\omega}{s^2+\omega^2}$. The solution in the s-domain is:

$$U(x,s) = \frac{\omega}{s^2+\omega^2} e^{-(x/c)s}$$

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To find the inverse transform, we use the Second Translation (t-Shifting) Theorem, which states $\mathscr{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)$, where H is the Heaviside step function. Here, $a = x/c$ and $F(s) = \frac{\omega}{s^2+\omega^2}$, so $f(t) = \sin(\omega t)$. Applying the theorem, the solution is:

$$u(x,t) = \sin\left(\omega\left(t - \frac{x}{c}\right)\right) H\left(t - \frac{x}{c}\right)$$

This represents a sinusoidal wave traveling to the right with speed c. The Heaviside function indicates that the displacement at a point x remains zero until time $t = x/c$, the time required for the wave to propagate from the origin to that point. ■

Finding Laplace Transforms

Find the Laplace transform, $F(s) = \mathscr{L}\{f(t)\}$, for each of the following functions.

1. $f(t) = 2t^3 - 4e^{-t} + 3\sin(5t)$
2. $f(t) = t^2 e^{3t}$
3. $f(t) = e^{-4t}\cosh(2t)$
4. $f(t) = \sin(t)\cos(t)$ (Hint: Use a trigonometric identity first.)
5. $f(t) = \frac{1-e^{-t}}{t}$ (Hint: Consider the transform of an integral.)
6. $f(t) = t^2\cos(3t)$ (Hint: Use the property for multiplication by t^n.)
7. $f(t)$ is a periodic square wave with period $T = 2$ defined by:
$$f(t) = \begin{cases} 1, & 0 \le t < 1 \\ -1, & 1 \le t < 2 \end{cases}$$
8. $f(t) = 5\delta(t) + t^3\delta(t-2)$

Finding Inverse Laplace Transforms

Find the inverse Laplace transform, $f(t) = \mathscr{L}^{-1}\{F(s)\}$, for each of the following functions.

9. $F(s) = \frac{3s+7}{s^2-2s-3}$
10. $F(s) = \frac{s-1}{s^2(s+3)}$
11. $F(s) = \frac{3s-2}{s^2+2s+10}$
12. $F(s) = \frac{s}{(s^2+4)^2}$ (Use the Convolution Theorem.)
13. $F(s) = \frac{e^{-2s}}{s^2+9}$
14. $F(s) = \ln\left(1 + \frac{\omega^2}{s^2}\right)$ (Hint: Differentiate with respect to s first.)

Solve Ordinary Differential Equations

Use the Laplace transform to solve the following initial value problems.

15. $y'' - 6y' + 9y = t^2e^{3t}$, with $y(0) = 2$ and $y'(0) = 6$.
16. $y'' + 4y = \sin(2t)$, with $y(0) = 1$ and $y'(0) = -1$.
17. $y'' - y' - 2y = 4t^2$, with $y(0) = 1$ and $y'(0) = 4$.
18. $y'(t) + \int_0^t y(\tau)d\tau = \cos(t)$, with $y(0) = 1$.

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Solve Partial Differential Equations

Use the Laplace transform to solve the following boundary value problems.

19. The one-dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = 9\frac{\partial^2 u}{\partial x^2}$ for $x > 0, t > 0$, subject to the conditions:
    - $u(x,0) = 0$ and $u_t(x,0) = 0$
    - $u(0,t) = t$ for $t \geq 0$
    - $\lim_{x\to\infty} u(x,t) = 0$
20. The one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ for a semi-infinite rod ($x > 0, t > 0$), subject to the conditions:
    - $u(x,0) = 50$
    - $u(0,t) = 0$ for $t > 0$
    - $u(x,t)$ is bounded as $x \to \infty$.

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CHAPTER

4

Tutorial - Integral Transforms

4.1 Applications to Differential and Integral Equations

Example 4.1.1 (Second-Order Homogeneous ODE)**.** Solve the initial value problem $y'' - y' - 6y = 0$ with initial conditions $y(0) = 1$ and $y'(0) = -1$.

Solution. Let $Y(s) = \mathscr{L}\{y(t)\}$. Taking the Laplace transform of the equation and incorporating the initial conditions, we get:

$$[s^2Y(s) - sy(0) - y'(0)] - [sY(s) - y(0)] - 6Y(s) = 0$$

$$s^2Y(s) - s(1) - (-1) - sY(s) + 1 - 6Y(s) = 0$$

We group terms to solve for $Y(s)$:

$$Y(s)(s^2 - s - 6) = s - 2$$

$$Y(s) = \frac{s-2}{s^2 - s - 6} = \frac{s-2}{(s-3)(s+2)}$$

Using partial fraction decomposition:

$$\frac{s-2}{(s-3)(s+2)} = \frac{A}{s-3} + \frac{B}{s+2}$$

The constants are found to be $A = \frac{3-2}{3+2} = \frac{1}{5}$ and $B = \frac{-2-2}{-2-3} = \frac{4}{5}$. So, $Y(s) = \frac{1}{5}\frac{1}{s-3} + \frac{4}{5}\frac{1}{s+2}$. Finding the inverse Laplace transform gives the solution:

$$y(t) = \frac{1}{5}\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} + \frac{4}{5}\mathscr{L}^{-1}\left\{\frac{1}{s+2}\right\} = \frac{1}{5}e^{3t} + \frac{4}{5}e^{-2t}$$

■

Example 4.1.2 (Second-Order Non-Homogeneous ODE)**.** Solve the initial value problem $y'' + 4y = 8t$ with $y(0) = 1$ and $y'(0) = 0$.

Solution. Transforming the equation gives:

$$[s^2Y(s) - sy(0) - y'(0)] + 4Y(s) = \mathscr{L}\{8t\}$$

$$s^2Y(s) - s(1) - 0 + 4Y(s) = \frac{8}{s^2}$$

Solving for $Y(s)$:

$$Y(s)(s^2 + 4) = s + \frac{8}{s^2} \implies Y(s) = \frac{s}{s^2+4} + \frac{8}{s^2(s^2+4)}$$

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We decompose the second term: $\frac{8}{s^2(s^2+4)} = \frac{2}{s^2} - \frac{2}{s^2+4}$.

$$Y(s) = \frac{s}{s^2+4} + \frac{2}{s^2} - \frac{2}{s^2+4}$$

Taking the inverse transform of each term:

$$\begin{aligned} y(t) &= \mathscr{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\} + 2\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} - \mathscr{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} \\ &= \cos(2t) + 2t - \sin(2t) \end{aligned}$$

■

Example 4.1.3 (System of First-Order ODEs)**.** Solve the system of differential equations for $t \geq 0$:

$$\begin{aligned} x'(t) &= -y(t) \\ y'(t) &= x(t) \end{aligned}$$

with initial conditions $x(0) = 1$ and $y(0) = 0$.

Solution. Let $X(s) = \mathscr{L}\{x(t)\}$ and $Y(s) = \mathscr{L}\{y(t)\}$. Transforming the system gives:

$$\begin{aligned} sX(s) - x(0) = -Y(s) &\implies sX(s) - 1 = -Y(s) \\ sY(s) - y(0) = X(s) &\implies sY(s) = X(s) \end{aligned}$$

This is a system of algebraic equations. Substituting the second equation into the first:

$$s(sY(s)) - 1 = -Y(s) \implies s^2Y(s) + Y(s) = 1 \implies Y(s)(s^2+1) = 1$$

$$Y(s) = \frac{1}{s^2+1}$$

Substituting this back to find $X(s)$:

$$X(s) = sY(s) = \frac{s}{s^2+1}$$

Finding the inverse transforms gives the solution pair:

$$y(t) = \mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin(t)$$

$$x(t) = \mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos(t)$$

■

Example 4.1.4 (Volterra Integral Equation with Convolution)**.** Solve the integral equation $y(t) = t + \int_0^t y(\tau)\sin(t-\tau)d\tau$.

Solution. We recognize the integral as the convolution $(y * \sin)(t)$. The equation is $y(t) = t + (y * \sin)(t)$. Taking the Laplace transform of the entire equation:

$$\mathscr{L}\{y(t)\} = \mathscr{L}\{t\} + \mathscr{L}\{(y * \sin)(t)\}$$

Using the Convolution Theorem, $\mathscr{L}\{(f * g)(t)\} = F(s)G(s)$:

$$Y(s) = \frac{1}{s^2} + Y(s) \cdot \mathscr{L}\{\sin(t)\} = \frac{1}{s^2} + Y(s)\frac{1}{s^2+1}$$

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Solving for $Y(s)$:

$$Y(s) - Y(s)\frac{1}{s^2+1} = \frac{1}{s^2} \implies Y(s)\left(1 - \frac{1}{s^2+1}\right) = \frac{1}{s^2}$$

$$Y(s)\left(\frac{s^2+1-1}{s^2+1}\right) = \frac{1}{s^2} \implies Y(s)\frac{s^2}{s^2+1} = \frac{1}{s^2}$$

$$Y(s) = \frac{s^2+1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}$$

Taking the inverse transform, we recall $\mathscr{L}\{t^n\} = n!/s^{n+1}$:

$$y(t) = \mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{3!}\mathscr{L}^{-1}\left\{\frac{3!}{s^4}\right\} = t + \frac{1}{6}t^3$$

■

Example 4.1.5 (Integro-Differential Equation)**.** Solve the integro-differential equation $y' + 2y + 5\int_0^t y(\tau)d\tau = 1$ with the initial condition $y(0) = 0$.

Solution. Taking the Laplace transform of each term:

$$\mathscr{L}\{y'\} + 2\mathscr{L}\{y\} + 5\mathscr{L}\left\{\int_0^t y(\tau)d\tau\right\} = \mathscr{L}\{1\}$$

$$[sY(s) - y(0)] + 2Y(s) + 5\frac{Y(s)}{s} = \frac{1}{s}$$

Substituting $y(0) = 0$ and multiplying the entire equation by s to clear the fraction:

$$s^2Y(s) + 2sY(s) + 5Y(s) = 1$$

$$Y(s)(s^2 + 2s + 5) = 1 \implies Y(s) = \frac{1}{s^2+2s+5}$$

To find the inverse transform, we complete the square in the denominator:

$$s^2 + 2s + 5 = (s^2 + 2s + 1) + 4 = (s+1)^2 + 2^2$$

So, $Y(s) = \frac{1}{(s+1)^2+2^2}$. We adjust this to match the sine transform form $\frac{b}{(s-a)^2+b^2}$:

$$Y(s) = \frac{1}{2} \cdot \frac{2}{(s+1)^2 + 2^2}$$

The inverse transform is found using the s-Shifting Theorem with $a = -1$ and $b = 2$:

$$y(t) = \frac{1}{2}e^{-t}\sin(2t)$$

■

Example 4.1.6 (ODE with a Dirac Delta Forcing Function)**.** Solve the initial value problem $y'' + 9y = \delta(t-2)$, with $y(0) = 0$ and $y'(0) = 0$.

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Solution. This models an oscillator at rest that receives a sharp "kick" or impulse at time $t = 2$. Transforming the equation:

$$[s^2Y(s) - sy(0) - y'(0)] + 9Y(s) = \mathscr{L}\{\delta(t-2)\}$$

$$s^2Y(s) + 9Y(s) = e^{-2s}$$

Solving for $Y(s)$:

$$Y(s)(s^2+9) = e^{-2s} \implies Y(s) = e^{-2s}\frac{1}{s^2+9}$$

We find the inverse transform using the Second Translation (t-Shifting) Theorem, which states that $\mathscr{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)$. Here, $a = 2$ and $F(s) = \frac{1}{s^2+9} = \frac{1}{3}\frac{3}{s^2+3^2}$. The inverse of $F(s)$ is $f(t) = \frac{1}{3}\sin(3t)$. Applying the theorem, the solution is:

$$y(t) = f(t-2)H(t-2) = \frac{1}{3}\sin(3(t-2))H(t-2)$$

This means the oscillator remains at rest until $t = 2$, after which it oscillates with an amplitude of $1/3$. ■

4.2 Applications to Boundary Value Problems

Example 4.2.1 (Heat Conduction with Initial Temperature)**.** A semi-infinite rod ($x \geq 0$) has a uniform initial temperature of u_0. At time $t = 0$, the end at $x = 0$ is instantaneously cooled to and maintained at a temperature of 0. Find the temperature $u(x,t)$ in the rod for $t > 0$.

Solution. The problem is described by the one-dimensional heat equation and its associated conditions:

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- Initial Condition (IC): $u(x,0) = u_0$
- Boundary Conditions (BCs): $u(0,t) = 0$, and $u(x,t)$ remains bounded as $x \to \infty$.

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. We transform the PDE with respect to time t:

$$sU(x,s) - u(x,0) = k\frac{d^2U}{dx^2} \implies sU(x,s) - u_0 = k\frac{d^2U}{dx^2}$$

This rearranges into a non-homogeneous second-order ODE for $U(x,s)$:

$$\frac{d^2U}{dx^2} - \frac{s}{k}U(x,s) = -\frac{u_0}{k}$$

The general solution is the sum of the complementary function and a particular solution. The particular solution is constant, $U_p = \frac{u_0}{s}$. The complementary solution is $U_c = C_1e^{\sqrt{s/k}x} + C_2e^{-\sqrt{s/k}x}$. Thus, the general solution is $U(x,s) = C_1e^{\sqrt{s/k}x} + C_2e^{-\sqrt{s/k}x} + \frac{u_0}{s}$. We now apply the transformed boundary conditions: $\mathscr{L}\{u(0,t)\} = U(0,s) = 0$ and $\lim_{x\to\infty} U(x,s)$ must be bounded.

- For $U(x,s)$ to be bounded as $x \to \infty$, the coefficient of the growing exponential term must be zero, so $C_1 = 0$.
- At $x = 0$, we have $U(0,s) = C_2 + \frac{u_0}{s} = 0$, which implies $C_2 = -\frac{u_0}{s}$.

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The solution in the s-domain is:

$$U(x,s) = \frac{u_0}{s} - \frac{u_0}{s} e^{-\frac{x}{\sqrt{k}}\sqrt{s}}$$

We find the inverse transform term-by-term. We know $\mathscr{L}^{-1}\{u_0/s\} = u_0$ and from standard tables, $\mathscr{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s}\right\} = \operatorname{erfc}\left(\frac{a}{2\sqrt{t}}\right)$, where erfc is the complementary error function. Here, $a = x/\sqrt{k}$. The inverse transform is:

$$u(x,t) = u_0 - u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) = u_0\left[1 - \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right)\right]$$

Since the error function $\operatorname{erf}(z) = 1 - \operatorname{erfc}(z)$, the solution is elegantly written as:

$$u(x,t) = u_0 \operatorname{erf}\left(\frac{x}{2\sqrt{kt}}\right)$$

■

Example 4.2.2 (Wave Propagation with a Step Input)**.** A semi-infinite elastic string ($x \geq 0$), initially at rest and in its equilibrium position, has its end at $x = 0$ abruptly displaced by a constant amount u_0 and held there. Find the displacement $u(x,t)$.

Solution. The governing equation is the one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- ICs: $u(x,0) = 0$ and $u_t(x,0) = 0$ (initially at rest)
- BCs: $u(0,t) = u_0$ for $t \geq 0$, and $u(x,t)$ is bounded as $x \to \infty$.

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. Transforming the PDE with respect to t:

$$s^2 U(x,s) - su(x,0) - u_t(x,0) = c^2 \frac{d^2U}{dx^2}$$

Substituting the zero initial conditions simplifies this to a homogeneous ODE in x:

$$\frac{d^2U}{dx^2} - \left(\frac{s}{c}\right)^2 U = 0$$

The general solution is $U(x,s) = C_1 e^{(s/c)x} + C_2 e^{-(s/c)x}$. The transformed boundary conditions are $U(0,s) = \mathscr{L}\{u_0\} = u_0/s$ and $\lim_{x\to\infty} U(x,s)$ is bounded.

- Boundedness requires $C_1 = 0$.
- At $x = 0$, $U(0,s) = C_2 = u_0/s$.

The solution in the s-domain is:

$$U(x,s) = \frac{u_0}{s} e^{-(x/c)s}$$

This form is a classic application of the Second Translation (t-Shifting) Theorem: $\mathscr{L}^{-1}\{e^{-as}F(s)\} = f(t-a)H(t-a)$. Here, $a = x/c$ and $F(s) = u_0/s$, so $f(t) = u_0$. The inverse transform is:

$$u(x,t) = u_0 H\left(t - \frac{x}{c}\right)$$

where H is the Heaviside step function. This solution represents a wavefront of height u_0 traveling to the right at speed c. The string at position x remains at rest until the wave reaches it at time $t = x/c$. ■

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Example 4.2.3 (Voltage in a Lossless Transmission Line)**.** A semi-infinite, lossless transmission line ($x \geq 0$) is initially uncharged. At $t = 0$, a constant voltage V_0 is applied at the sending end ($x = 0$). Find the voltage $v(x,t)$ along the line.

Solution. The voltage on a lossless line is governed by the wave equation, where the wave speed is $c = 1/\sqrt{LC}$ (L and C are inductance and capacitance per unit length).

$$\frac{\partial^2 v}{\partial t^2} = c^2 \frac{\partial^2 v}{\partial x^2}, \quad x > 0, t > 0$$

- ICs: $v(x,0) = 0$ and $v_t(x,0) = 0$ (initially uncharged and no current)
- BCs: $v(0,t) = V_0$ for $t \geq 0$, and $v(x,t)$ is bounded as $x \to \infty$.

This problem is mathematically identical to the previous example (Example 2). Following the exact same procedure, we transform the PDE and boundary conditions, solve the resulting ODE for the transformed voltage $V(x,s)$, and find:

$$V(x,s) = \frac{V_0}{s} e^{-(x/c)s}$$

Taking the inverse Laplace transform gives the voltage along the line:

$$v(x,t) = V_0 H\left(t - \frac{x}{c}\right)$$

This describes a step voltage of magnitude V_0 propagating down the transmission line at speed c. ■

Example 4.2.4 (Heat Conduction with a Ramp Temperature at the Boundary)**.** A semi-infinite rod ($x \geq 0$) is initially at zero temperature. At $t = 0$, the temperature at the end $x = 0$ is increased linearly with time, such that $u(0,t) = At$ for some constant A. Find the temperature $u(x,t)$.

Solution. The problem setup is:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0$$

- IC: $u(x,0) = 0$
- BCs: $u(0,t) = At$, and $u(x,t)$ is bounded as $x \to \infty$.

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. Transforming the PDE and using the zero initial condition gives the familiar ODE:

$$\frac{d^2 U}{dx^2} - \frac{s}{k} U(x,s) = 0$$

The general solution that remains bounded as $x \to \infty$ is $U(x,s) = Ce^{-\sqrt{s/k}x}$. The transformed boundary condition at $x = 0$ is $U(0,s) = \mathscr{L}\{At\} = A/s^2$. Applying this condition gives $C = A/s^2$. The solution in the s-domain is:

$$U(x,s) = \frac{A}{s^2} e^{-\frac{x}{\sqrt{k}}\sqrt{s}}$$

Finding the inverse of this expression is more advanced and requires tables of transform pairs. A standard pair is:

$$\mathscr{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s^2}\right\} = (t + \frac{a^2}{2})\text{erfc}\left(\frac{a}{2\sqrt{t}}\right) - a\sqrt{\frac{t}{\pi}} e^{-a^2/4t}$$

Using $a = x/\sqrt{k}$, we obtain the temperature distribution:

$$u(x,t) = A\left[\left(t + \frac{x^2}{2k}\right)\operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right) - x\sqrt{\frac{t}{k\pi}}e^{-x^2/4kt}\right]$$

This demonstrates how the Laplace transform method can handle more complex boundary conditions, even if the final inversion step relies on known, non-trivial transform pairs. ■

4.3 Applications in Engineering and Physics

Example 4.3.1 (Electrical Circuits: RLC Series Circuit)**.** An RLC series circuit has an inductor of $L = 1$ H, a resistor of $R = 20\ \Omega$, a capacitor of $C = 0.01$ F, and a constant voltage source of $E = 12$ V. The circuit is switched on at $t = 0$. Find the current $i(t)$ in the circuit, assuming the initial current and initial charge on the capacitor are both zero.

Solution. The governing equation for an RLC circuit from Kirchhoff's Voltage Law is:

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int_0^t i(\tau)d\tau = E(t)$$

Substituting the given values:

$$\frac{di}{dt} + 20i + 100\int_0^t i(\tau)d\tau = 12$$

The initial condition is $i(0) = 0$. Let $I(s) = \mathscr{L}\{i(t)\}$. We take the Laplace transform of the equation:

$$[sI(s) - i(0)] + 20I(s) + 100\frac{I(s)}{s} = \frac{12}{s}$$

Substituting $i(0) = 0$ and multiplying by s to clear the fractions:

$$s^2I(s) + 20sI(s) + 100I(s) = 12$$

Solving for $I(s)$:

$$I(s)(s^2 + 20s + 100) = 12 \implies I(s) = \frac{12}{(s+10)^2}$$

To find the inverse transform, we use the s-Shifting Theorem. We know that $\mathscr{L}^{-1}\{1/s^2\} = t$.

$$i(t) = 12\mathscr{L}^{-1}\left\{\frac{1}{(s+10)^2}\right\} = 12e^{-10t}\mathscr{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$i(t) = 12te^{-10t} \text{ Amperes}$$

This represents a critically damped response where the current rises to a peak and then decays to zero. ■

Example 4.3.2 (Dynamics: Damped Forced Spring-Mass System)**.** A mass of $m = 1$ kg is attached to a spring with constant $k = 5$ N/m. The system is damped with a damping coefficient of $b = 4$ Ns/m. The mass is initially at rest at its equilibrium position. At $t = 0$, it is subjected to an external force $F(t) = e^{-2t}$ N. Find the displacement $x(t)$ of the mass.

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Solution. The equation of motion for a spring-mass-damper system is $mx'' + bx' + kx = F(t)$. Substituting the given values:

$$x'' + 4x' + 5x = e^{-2t}$$

The initial conditions are $x(0) = 0$ and $x'(0) = 0$. Let $X(s) = \mathscr{L}\{x(t)\}$. Transforming the equation:

$$[s^2X(s) - sx(0) - x'(0)] + 4[sX(s) - x(0)] + 5X(s) = \mathscr{L}\{e^{-2t}\}$$

$$s^2X(s) + 4sX(s) + 5X(s) = \frac{1}{s+2}$$

Solving for $X(s)$:

$$X(s)(s^2 + 4s + 5) = \frac{1}{s+2} \implies X(s) = \frac{1}{(s+2)(s^2+4s+5)}$$

We use partial fraction decomposition: $X(s) = \frac{A}{s+2} + \frac{Bs+C}{s^2+4s+5}$. Solving for the constants gives $A = 1$, $B = -1$, and $C = -2$.

$$X(s) = \frac{1}{s+2} - \frac{s+2}{s^2+4s+5}$$

To invert the second term, we complete the square: $s^2 + 4s + 5 = (s+2)^2 + 1^2$.

$$X(s) = \frac{1}{s+2} - \frac{s+2}{(s+2)^2+1^2}$$

We now find the inverse transform of each term using the s-Shifting Theorem:

$$x(t) = \mathscr{L}^{-1}\left\{\frac{1}{s+2}\right\} - \mathscr{L}^{-1}\left\{\frac{s+2}{(s+2)^2+1^2}\right\}$$

$$x(t) = e^{-2t} - e^{-2t}\cos(t) \text{ meters}$$

■

Example 4.3.3 (Beams: Cantilever Beam with a Point Load)**.** A cantilever beam of length L and flexural rigidity EI is fixed at $x = 0$. It is subjected to a single concentrated downward load P at its free end, $x = L$. Find the deflection $y(x)$ of the beam.

Solution. The governing equation for beam deflection is $EI\frac{d^4y}{dx^4} = w(x)$, where $w(x)$ is the distributed load. A point load can be represented using the Dirac delta function: $w(x) = P\delta(x - L)$. The equation becomes:

$$EI\frac{d^4y}{dx^4} = P\delta(x - L)$$

The boundary conditions for a cantilever beam are $y(0) = 0$ (zero deflection at the fixed end) and $y'(0) = 0$ (zero slope at the fixed end). We use the Laplace transform with respect to the spatial variable x. Let $Y(s) = \mathscr{L}\{y(x)\}$. We also need to define the unknown shear and bending moment at the fixed end: $y''(0) = C_1$ and $y'''(0) = C_2$. Transforming the equation:

$$EI[s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)] = P\mathscr{L}\{\delta(x - L)\}$$

$$EI[s^4Y(s) - sC_1 - C_2] = Pe^{-Ls}$$

Solving for $Y(s)$:

$$Y(s) = \frac{C_1}{s^3} + \frac{C_2}{s^4} + \frac{P}{EI}\frac{e^{-Ls}}{s^4}$$

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The inverse transform gives: $y(x) = C_1\frac{x^2}{2} + C_2\frac{x^3}{6} + \frac{P}{EI}\frac{(x-L)^3}{6}H(x-L)$. To find C_1 and C_2, we use the boundary conditions at the free end ($x = L$): zero bending moment ($y''(L) = 0$) and zero shear force ($y'''(L) = 0$). $y''(x) = C_1 + C_2x + \frac{P}{EI}(x-L)H(x-L) \implies y''(L) = C_1 + C_2L = 0$. $y'''(x) = C_2 + \frac{P}{EI}H(x-L) \implies y'''(L) = C_2 + \frac{P}{EI} = 0$. From the second equation, $C_2 = -P/EI$. Substituting into the first gives $C_1 - PL/EI = 0 \implies C_1 = PL/EI$. Substituting the constants back into the solution for $y(x)$:

$$y(x) = \frac{PL}{EI}\frac{x^2}{2} - \frac{P}{EI}\frac{x^3}{6} = \frac{P}{6EI}(3Lx^2 - x^3) \quad \text{for } 0 \le x < L$$

■

Example 4.3.4 (Heat Conduction: Semi-Infinite Rod with Heat Flux)**.** A semi-infinite rod ($x \ge 0$) is initially at zero temperature. At $t = 0$, a constant heat flux q_0 is applied at the end $x = 0$. Find the temperature $u(x,t)$.

Solution. The problem is governed by the heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$. The boundary condition related to heat flux is given by Fourier's Law of Conduction, $q = -K\frac{\partial u}{\partial x}$, where K is thermal conductivity. So, at $x = 0$, we have $-K\frac{\partial u}{\partial x}(0,t) = q_0$. The conditions are:

- IC: $u(x,0) = 0$
- BCs: $\frac{\partial u}{\partial x}(0,t) = -\frac{q_0}{K}$, and $u(x,t)$ is bounded as $x \to \infty$.

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. The transformed PDE is $\frac{d^2U}{dx^2} - \frac{s}{k}U = 0$. The bounded solution is $U(x,s) = Ce^{-\sqrt{s/k}x}$. The transformed flux boundary condition is $\frac{dU}{dx}(0,s) = \mathscr{L}\{-\frac{q_0}{K}\} = -\frac{q_0}{Ks}$. We differentiate $U(x,s)$ with respect to x: $\frac{dU}{dx} = -C\sqrt{s/k}e^{-\sqrt{s/k}x}$. At $x = 0$: $\frac{dU}{dx}(0,s) = -C\sqrt{s/k} = -\frac{q_0}{Ks}$. Solving for C: $C = \frac{q_0\sqrt{k}}{K}\frac{1}{s\sqrt{s}} = \frac{q_0\sqrt{k}}{K}s^{-3/2}$. So, $U(x,s) = \frac{q_0\sqrt{k}}{K}\frac{e^{-\frac{x}{\sqrt{k}}\sqrt{s}}}{s^{3/2}}$. This is a known, non-trivial inverse transform pair: $\mathscr{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s^{3/2}}\right\} = 2\sqrt{\frac{t}{\pi}}e^{-a^2/4t} - a\,\text{erfc}(\frac{a}{2\sqrt{t}})$. With $a = x/\sqrt{k}$:

$$u(x,t) = \frac{q_0\sqrt{k}}{K}\left[2\sqrt{\frac{t}{\pi}}e^{-x^2/4kt} - \frac{x}{\sqrt{k}}\text{erfc}\left(\frac{x}{2\sqrt{kt}}\right)\right]$$

■

Example 4.3.5 (Wave Equation: Plucked String)**.** A semi-infinite string ($x \ge 0$) is lifted to an initial displacement $u(x,0) = Ae^{-bx}$ and released from rest ($u_t(x,0) = 0$). Find the subsequent motion $u(x,t)$.

Solution. The motion is governed by $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$.

- ICs: $u(x,0) = Ae^{-bx}$, $u_t(x,0) = 0$.
- BCs: $u(0,t)$ is bounded (implied, no explicit condition needed here for the method).

Let $U(x,s) = \mathscr{L}\{u(x,t)\}$. Transforming the PDE gives:

$$s^2U(x,s) - su(x,0) - u_t(x,0) = c^2\frac{d^2U}{dx^2}$$

$$s^2U - sAe^{-bx} = c^2\frac{d^2U}{dx^2} \implies \frac{d^2U}{dx^2} - \left(\frac{s}{c}\right)^2 U = -\frac{sA}{c^2}e^{-bx}$$

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This is a non-homogeneous ODE for $U(x,s)$. The particular solution has the form $U_p = Ce^{-bx}$. Substituting gives $C(b^2 - s^2/c^2) = -sA/c^2$, so $C = \frac{sA}{s^2 - b^2c^2}$. The bounded complementary solution is $U_c = De^{-(s/c)x}$. So, $U(x,s) = De^{-(s/c)x} + \frac{sA}{s^2-(bc)^2}e^{-bx}$. The constant D must be chosen so that the solution reflects the physical reality. A direct inversion of this form is complex. However, d'Alembert's solution for an infinite string is $u(x,t) = \frac{1}{2}[f(x-ct) + f(x+ct)]$. For a semi-infinite string fixed at $x = 0$, we use an odd extension of the initial shape. In this case, with a free end, we use an even extension. The solution is:

$$u(x,t) = \frac{A}{2}[e^{-b(x-ct)}H(x-ct) + e^{-b(x+ct)}]$$

This represents two exponential shapes, one traveling right and one traveling left (which reflects off the "virtual" string for $x < 0$). ■

Example 4.3.6 (Dynamics: Rotational Motion with Damping)**.** A flywheel with moment of inertia $J = 10$ kg·m^2 is subject to a viscous damping torque of $B = 20$ N·m·s/rad. It is initially at rest. At $t = 0$, a constant torque $\tau = 100$ N·m is applied. Find the angular velocity $\omega(t)$.

Solution. The equation for rotational motion is $J\frac{d\omega}{dt} + B\omega = \tau(t)$. Substituting the values:

$$10\frac{d\omega}{dt} + 20\omega = 100 \implies \frac{d\omega}{dt} + 2\omega = 10$$

The initial condition is $\omega(0) = 0$. Let $\Omega(s) = \mathscr{L}\{\omega(t)\}$. Transforming the equation:

$$[s\Omega(s) - \omega(0)] + 2\Omega(s) = \frac{10}{s}$$

$$s\Omega(s) + 2\Omega(s) = \frac{10}{s} \implies \Omega(s)(s+2) = \frac{10}{s}$$

$$\Omega(s) = \frac{10}{s(s+2)}$$

Using partial fractions, $\frac{10}{s(s+2)} = \frac{5}{s} - \frac{5}{s+2}$.

$$\Omega(s) = \frac{5}{s} - \frac{5}{s+2}$$

The inverse transform gives the angular velocity:

$$\omega(t) = \mathscr{L}^{-1}\left\{\frac{5}{s}\right\} - \mathscr{L}^{-1}\left\{\frac{5}{s+2}\right\} = 5 - 5e^{-2t} \text{ rad/s}$$

The flywheel's speed exponentially approaches a terminal velocity of 5 rad/s. ■

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A Concluding Note to Students

"Mathematics reveals order within complexity. Fourier analysis uncovers hidden frequencies, while integral transforms turn difficult problems into simpler mathematical representations."

DEAR STUDENTS,

Congratulations on completing Fourier Series and Integral Transforms!

Throughout this course, we have discovered how complicated functions and mathematical problems can be understood through simpler representations. Fourier series introduced us to the harmonic structure of periodic functions, while Fourier transforms revealed the relationship between functions and their frequency components. The discrete Fourier transform and fast Fourier transform connected these ideas with numerical computation and signal processing.

Laplace transforms provided another powerful approach, enabling us to solve differential equations and study problems arising in physics and engineering. Together, these techniques demonstrate how mathematical reasoning can transform difficult questions into manageable ones.

The true value of this course lies beyond memorizing formulas. It lies in understanding their meaning, recognizing when to use them, and developing the confidence to approach unfamiliar problems. As you continue your studies, remember that patience, consistent practice, and curiosity are essential to mathematical progress.

May these notes strengthen your foundations and encourage you to explore mathematics beyond examinations and the classroom.

Keep your curiosity alive! Every mathematical challenge is an opportunity to discover new ideas and develop deeper understanding.

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